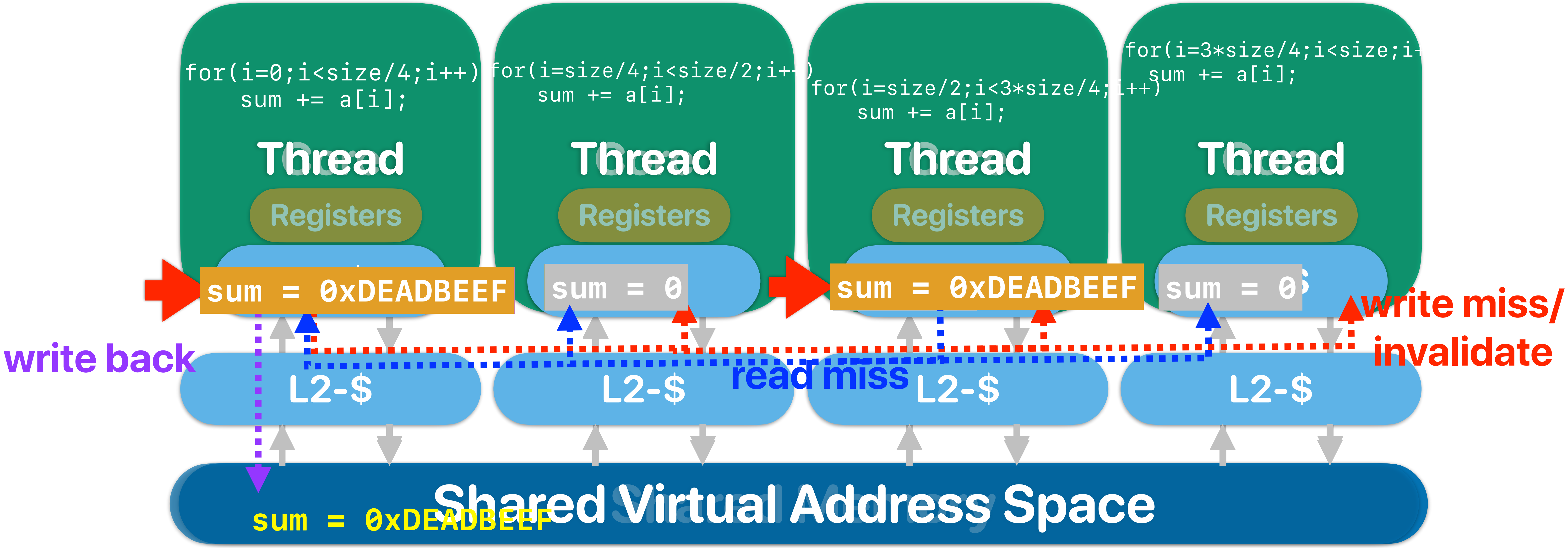
Parallel programming (cont.) & The dark silicon problem

Hung-Wei Tseng

Recap: Modern processors have both CMP/SMT

What happens when we write in coherent caches?



Cache coherency

- Assuming that we are running the following code on a CMP with a cache coherency protocol, how many of the following outputs are possible? (a is initialized to 0 as assume we will output more than 10 numbers)

thread 1	thread 2
`while(1)` `    printf("%d ",a);`	`while(1)` `    a++;`

① 0 1 2 3 4 5 6 7 8 9
② 1 2 5 9 3 6 8 10 12 13
③ 1 1 1 1 1 1 1 64 100
④ 1 1 1 1 1 1 1 1 1 100

A. 0
B. 1
C. 2
D. 3
E. 4

Recap: 4Cs of cache misses

- 3Cs:
  - Compulsory, Conflict, Capacity
- Coherency miss:
  - A "block" invalidated because of the sharing among processors.
- True sharing
  - Processor A modifies X, processor B also want to access X.
- False sharing
  - Processor A modifies X, processor B also want to access Y. However, Y is invalidated because X and Y are in the same block!

Again — how many values are possible?

- Consider the given program. You can safely assume the caches are coherent. How many of the following outputs will you see?

  ① (0, 0)
  ② (0, 1)
  ③ (1, 0)
  ④ (1, 1)

  A. 0
  B. 1
  C. 2
  D. 3
  E. 4

```c
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <unistd.h>

volatile int a,b;
volatile int x,y;
volatile int f;
void* modifya(void *z) {
  a=1;
  x=b;
  return NULL;
}
void* modifyb(void *z) {
  b=1;
  y=a;
  return NULL;
}
```

```c
int main() {
  int i;
  pthread_t thread[2];
  pthread_create(&thread[0], NULL, modifya, NULL);
  pthread_create(&thread[1], NULL, modifyb, NULL);
  pthread_join(thread[0], NULL);
  pthread_join(thread[1], NULL);
  fprintf(stderr,"(%d, %d)\n",x,y);
  return 0;
}
```

Possible scenarios

Thread 1	Thread 2
`a=1;`	
	`b=1;`
	`y=a;`
`x=b;`	

(1,1)

Thread 1	Thread 2
`a=1;`	
`x=b;`	
	`b=1;`
	`y=a;`

(0,1)

Thread 1	Thread 2
	`b=1;`
	`y=a;`
`a=1;`	
`x=b;`	

(1,0)

Thread 1	Thread 2
	`y=a;`
`x=b;`	
`a=1;`	
	`b=1;`

OoO Scheduling!

(0,0)

Why (0,0)?

- Processor/compiler may reorder your memory operations/ instructions
  - Coherence protocol can only guarantee the update of the same memory address
  - Processor can serve memory requests without cache miss first
  - Compiler may store values in registers and perform memory operations later
- Each processor core may not run at the same speed (cache misses, branch mis-prediction, I/O, voltage scaling and etc..)
- Threads may not be executed/scheduled right after it's spawned

Again — how many values are possible?

- Consider the given program. You can safely assume the caches are coherent. How many of the following outputs will you see?

  ① (0, 0)
  ② (0, 1)
  ③ (1, 0)
  ④ (1, 1)

  A. 0
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  return 0;
}
```

Take-aways: parallel programming

- Cache coherency only guarantees that everyone would eventually have a coherent view of data, but not when
- Cache coherency may create unexpected cache invalidations/misses if you do it wrong
- Processor behaviors are non-deterministic
  - You cannot predict which processor is going faster
  - You cannot predict when OS is going to schedule your thread
  - You cannot predict when the processor is going to schedule an instruction

fence instructions

- x86 provides an “mfence” instruction to prevent reordering across the fence instruction
  - All updates prior to mfence must finish before the instruction can proceed
- x86 only supports this kind of “relaxed consistency” model. You still have to be careful enough to make sure that your code behaves as you expected

thread 1	thread 2

`a=1;`
`mfence` **a=1 must occur/update before mfence**
`x=b;`
	`b=1;`
	`mfence` **b=1 must occur/update before mfence**
	`y=a;`

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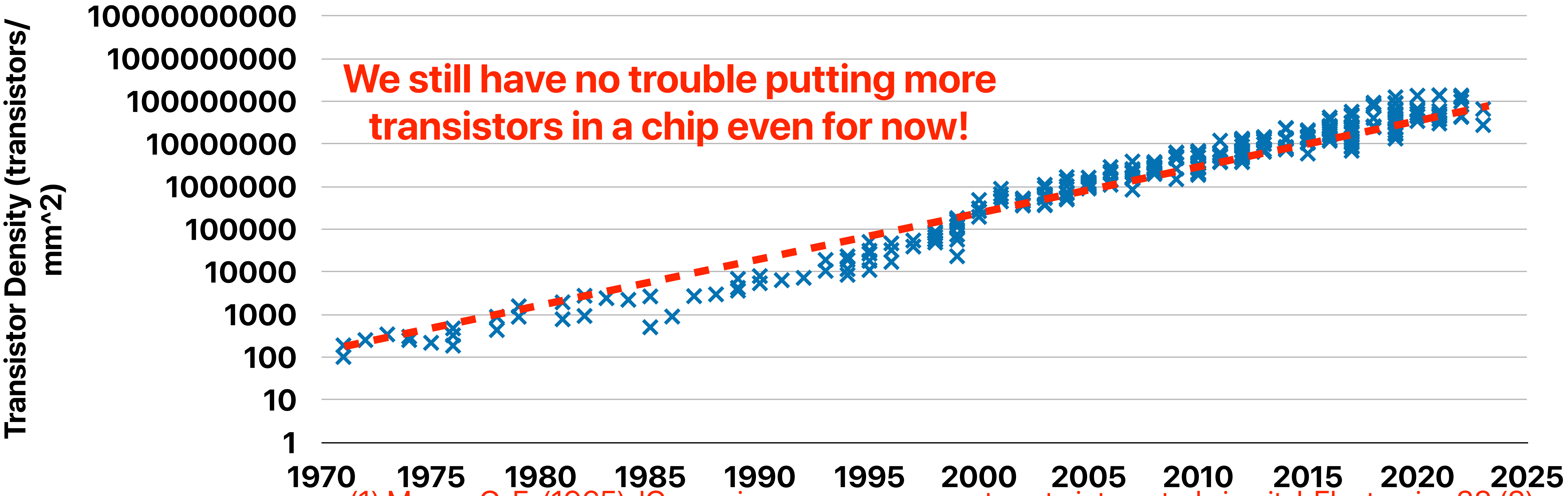
Outline

- The dark silicon problem
- State-of-the-art “work-arounds” in the dark silicon era

What is and why is dark silicon?

Moore's Law[1]

- The number of transistors we can build in a fixed area of silicon doubles every 12 ~ 24 months.
- Moore's Law "was" the most important driver for historic CPU performance gains



(1) Moore, G. E. (1965), 'Cramming more components onto integrated circuits', Electronics 38 (8) .

Plot based on https://en.wikipedia.org/wiki/Transistor_count by Hung-Wei Tseng

The broken Dennard Scaling

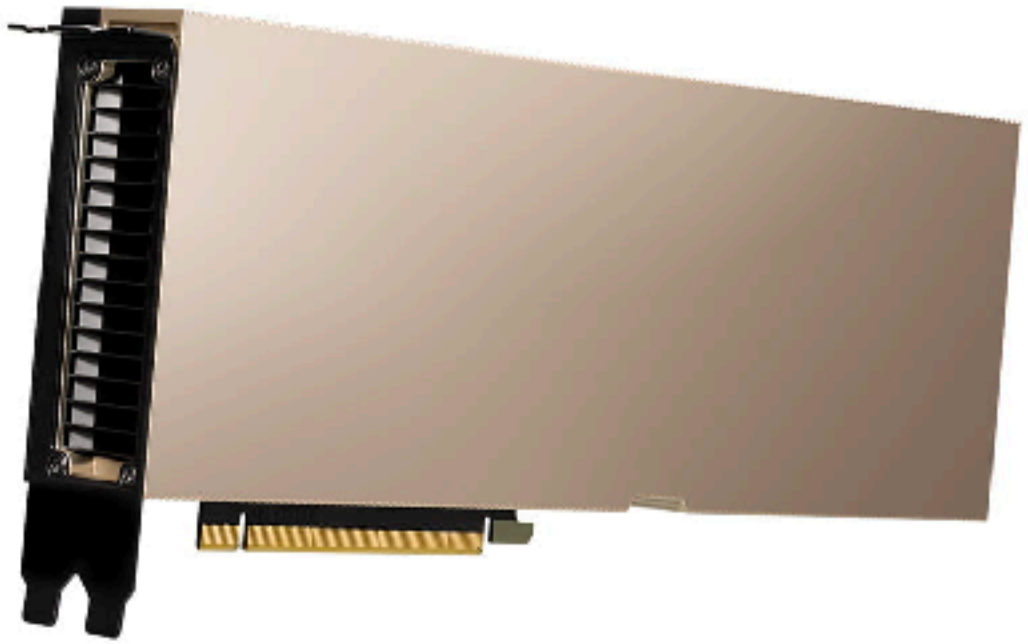
NVIDIA Accelerator Specification Comparison			
	H100	A100 (80GB)	V100
FP32 CUDA Cores	16896	6912	5120
Tensor Cores	528	432	640
GPU	GH100 (814mm2)	GA100 (826mm2)	GV100 (815mm2)
Transistor Count	80B	54.2B	21.1B
TDP	700W	400W	300W/350W
Manufacturing Process	TSMC 4N	TSMC 7N	TSMC 12nm FFN
Interface	SXM5	SXM4	SXM2/SXM3
Architecture	Hopper	Ampere	Volta

1.46x

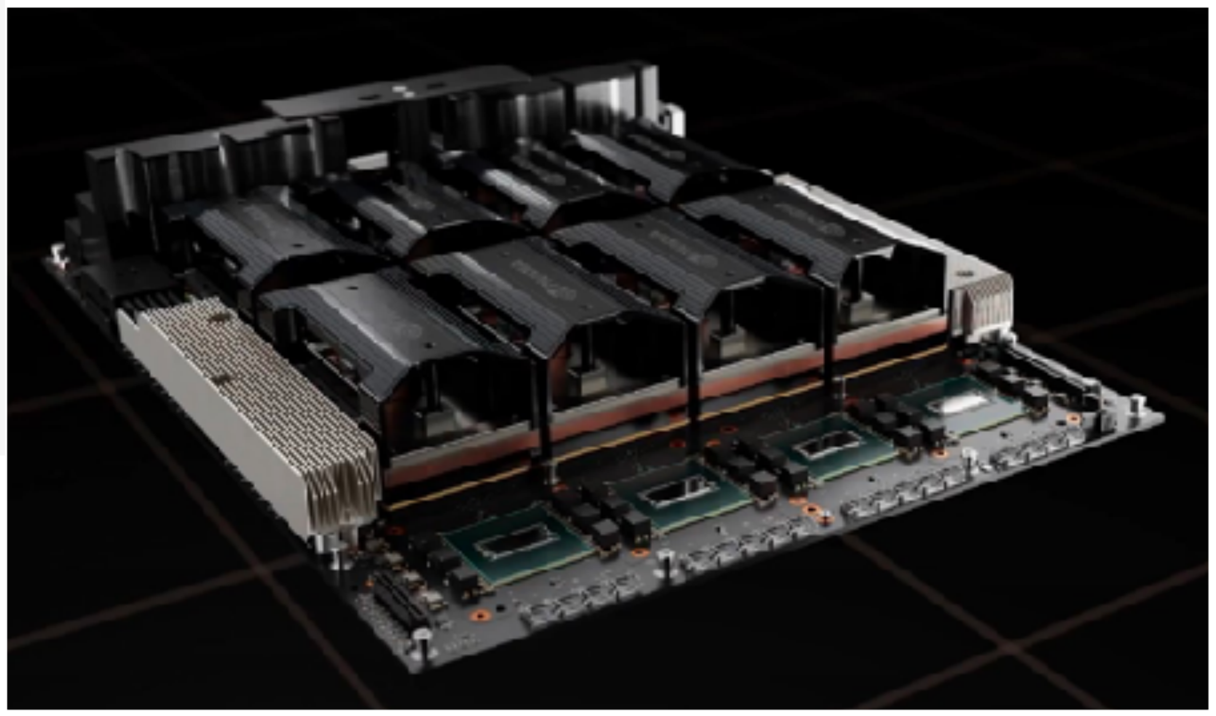
1.75x

8.75 W/ B Transistors

7.38 W/ B Transistors



https://www.workstationspecialist.com/product/nvidia-tesla-a100/



https://www.servethehome.com/wp-content/uploads/2022/03/NVIDIA-GTC-2022-H100-in-HGX-H100.jpg

https://www.pollev.com/hungweitseng close in 1:30

What happens if power doesn’t scale with process technologies?

- If we are able to cram more transistors within the same chip area (Moore’s law continues), but the power consumption per transistor remains the same. Right now, if put more transistors in the same area because the technology allows us to. How many of the following statements are true?
  ① The power consumption per chip will increase
  ② The power density of the chip will increase
  ③ Given the same power budget, we may not able to power on all chip area if we maintain the same clock rate
  ④ Given the same power budget, we may have to lower the clock rate of circuits to power on all chip area

  A. 0
  B. 1
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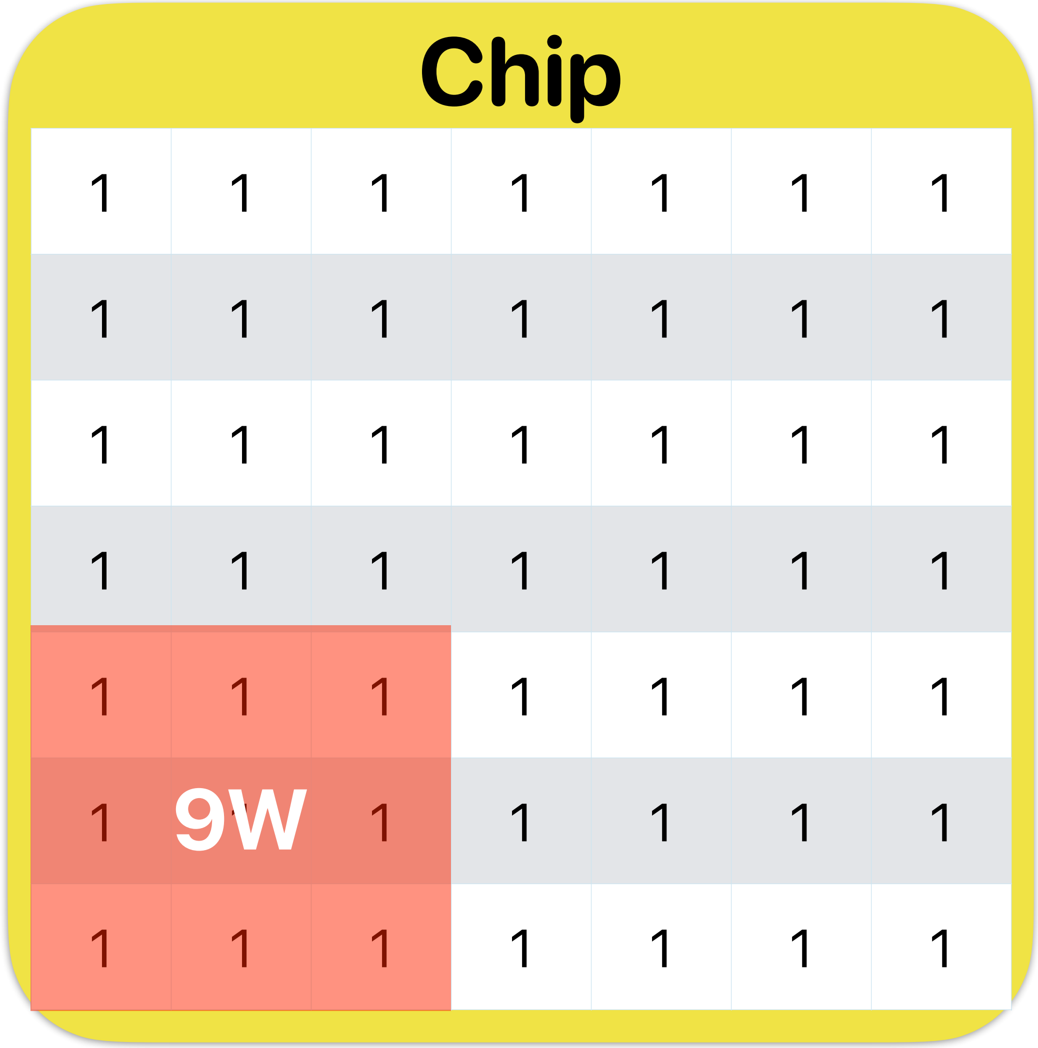
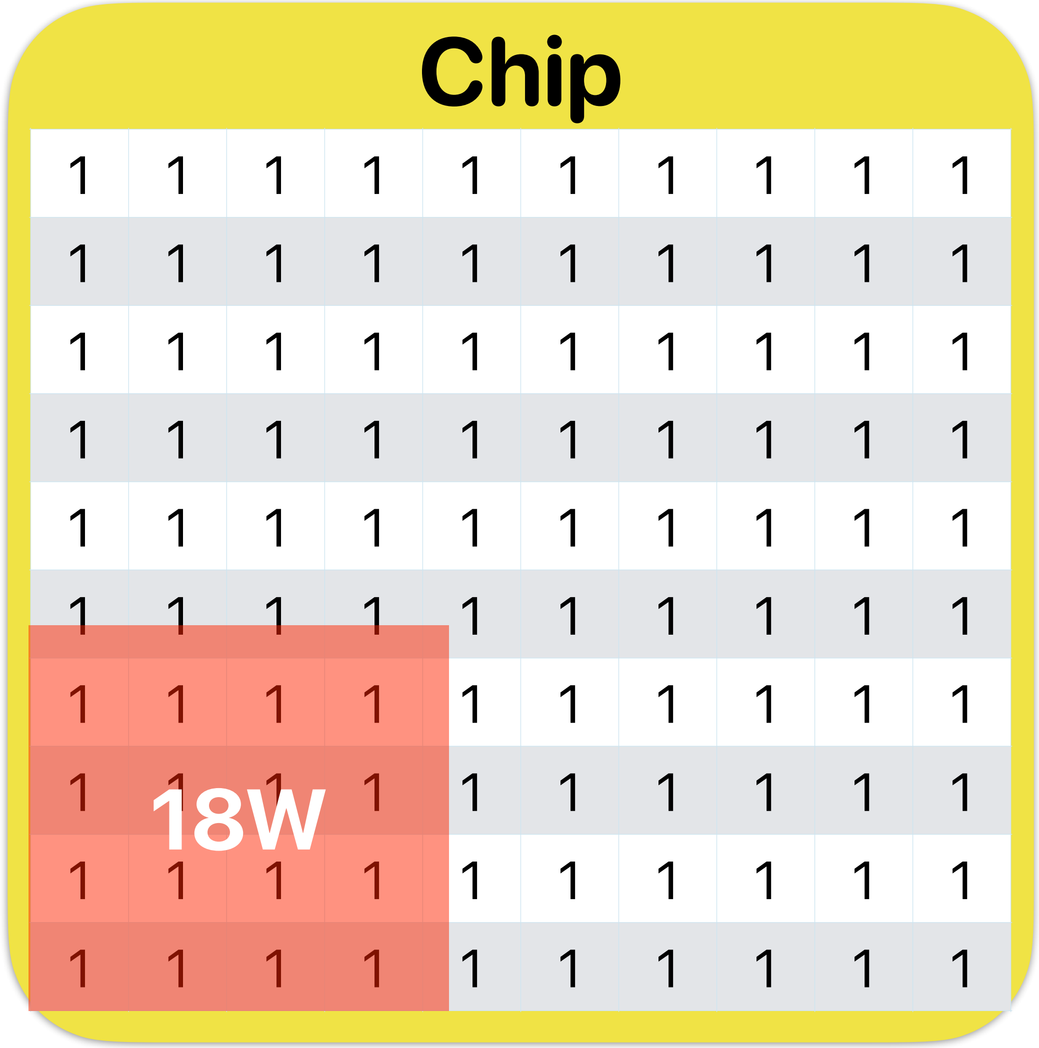
Power consumption to light on all transistors



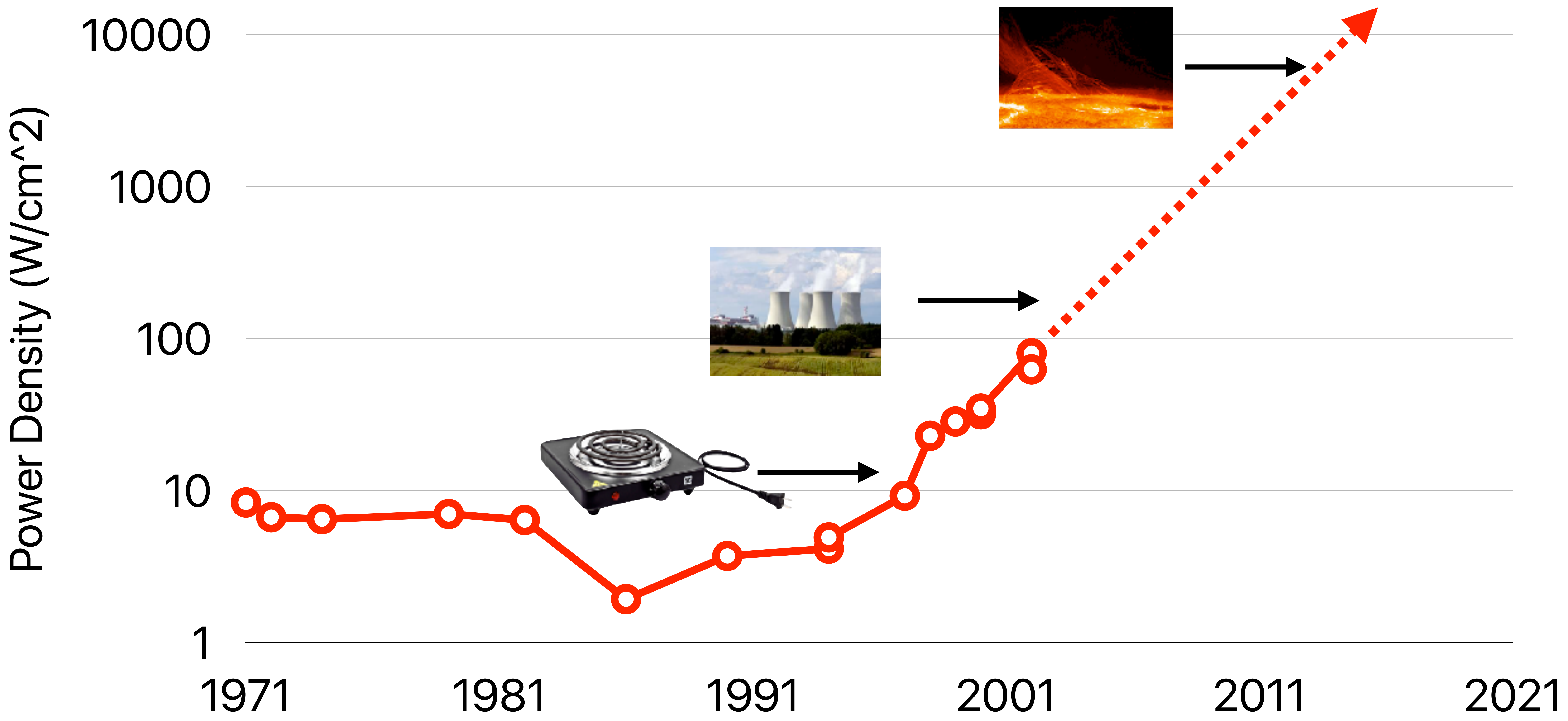
=49W

Dennardian Broken

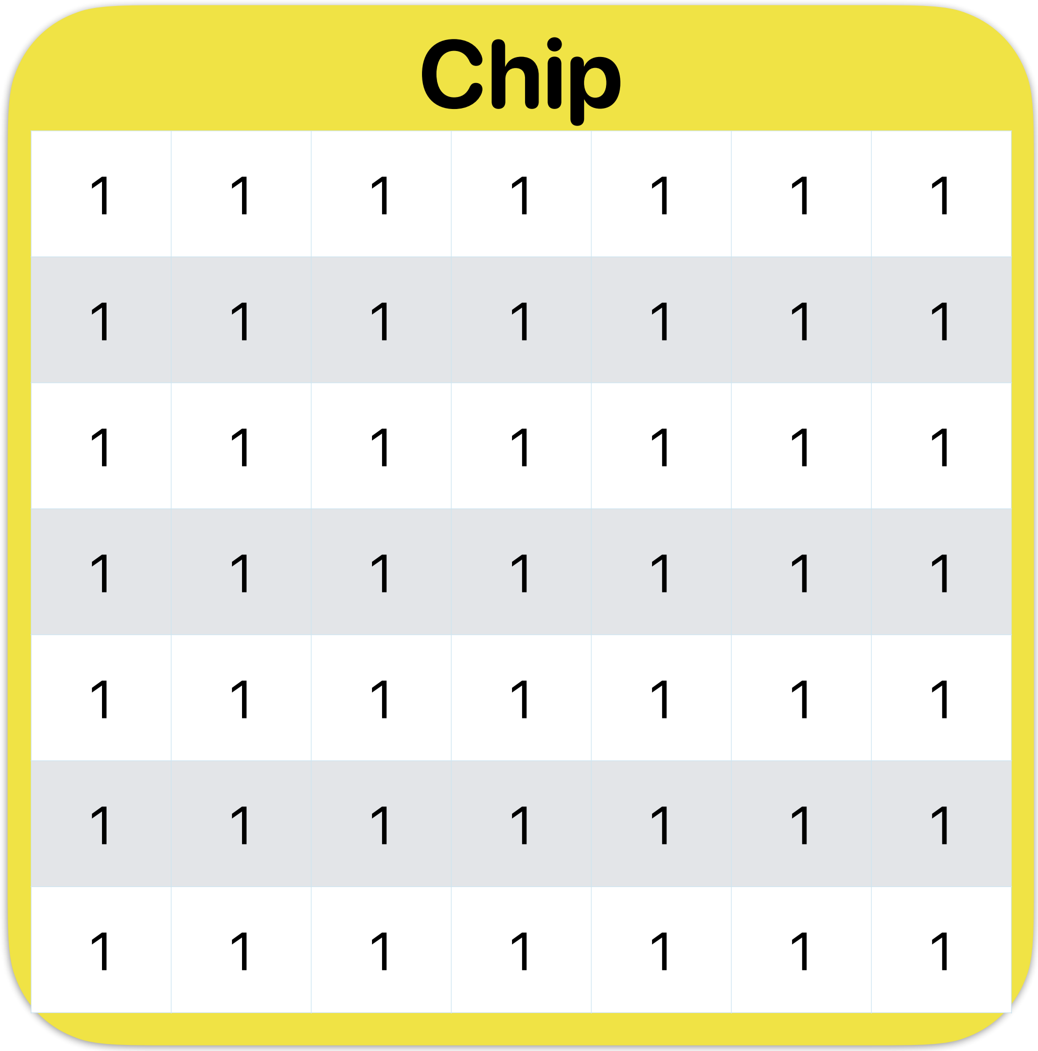


=100W!

Power Density of Processors

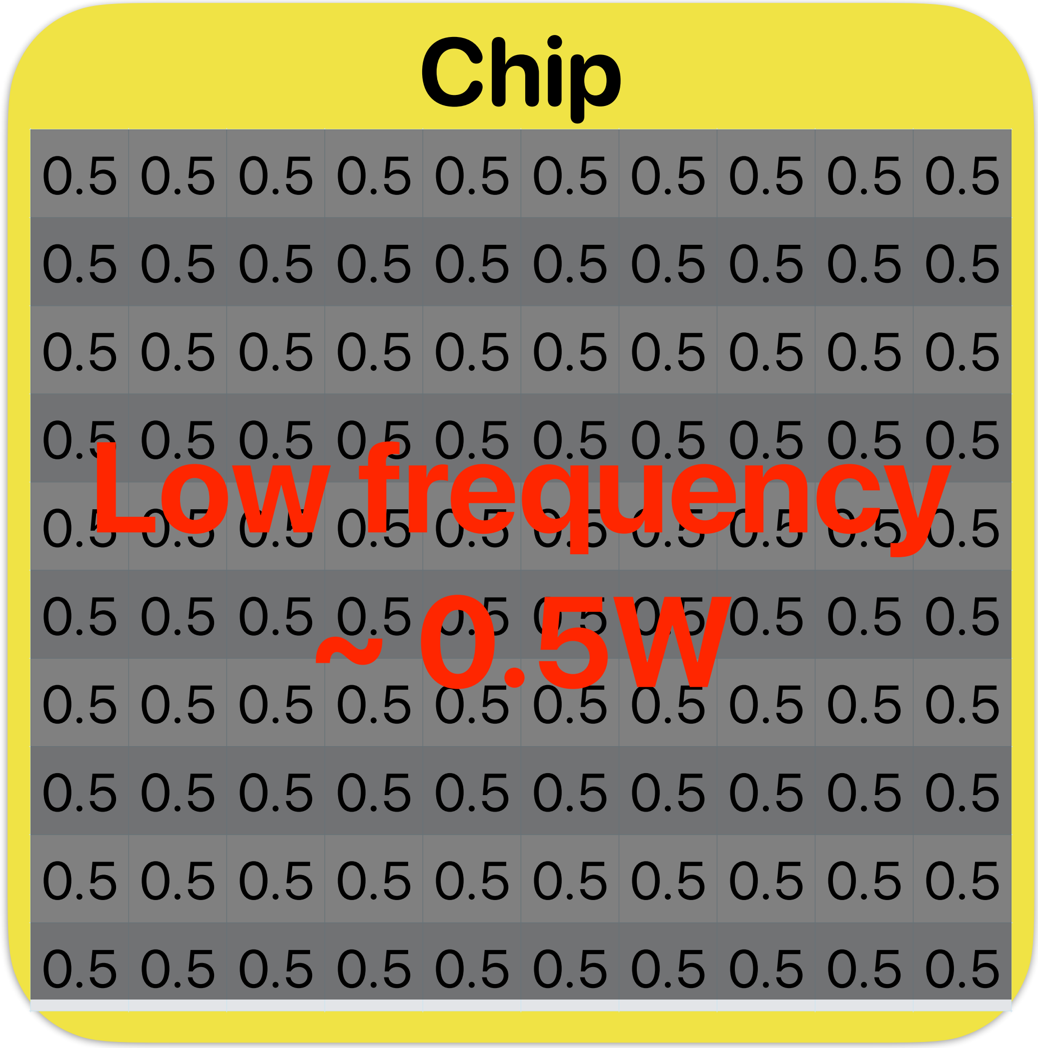


If we have a fixed power budget



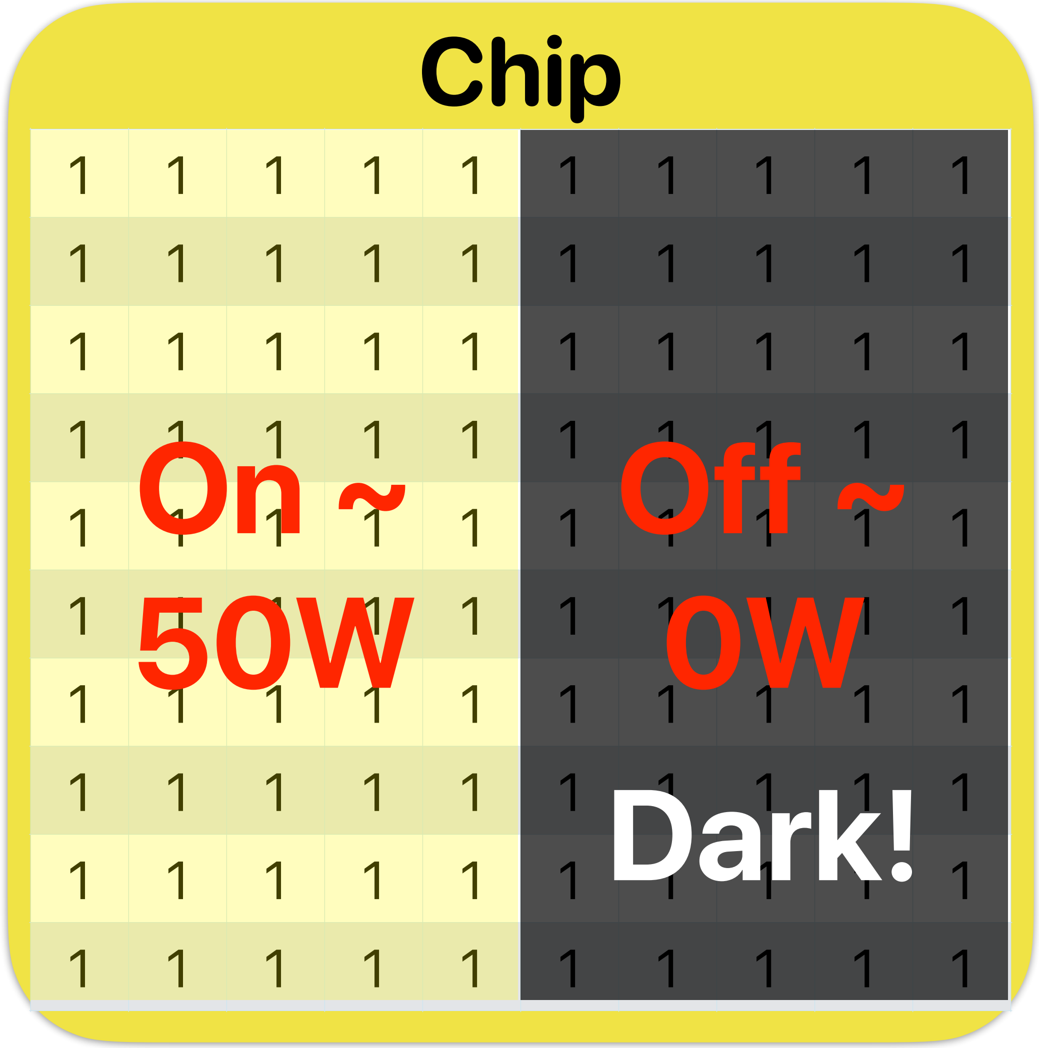
=49W

Slowdown all of them



=50W!

Dark silicon!



=50W!

Power consumption & power density

- $P_{dynamic} \sim \alpha \times C \times \boxed{V^2} \times f \times N$
  - α: average switches per cycle
  - C: capacitance
  - V: voltage
  - f: frequency, usually linear with V
  - N: the number of transistors
- $P_{leakage} \sim N \times \boxed{V} \times e^{-V_t}$
  - N: number of transistors
  - V: voltage
  - V_t: threshold voltage where transistor conducts (begins to switch)
- Power density:

$$P_{density} = \frac{P}{area}$$

Dennard scaling discontinued — we cannot make voltage lower

Moore's Law allows higher frequencies as transistors are smaller

Moore's Law makes this smaller

What happens if power doesn't scale with process technologies?

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  ① The power consumption per chip will increase
  ② The power density of the chip will increase
  ③ Given the same power budget, we may not able to power on all chip area if we maintain the same clock rate — **even if we put more cores, we cannot have all of them on!**
  ④ Given the same power budget, we may have to lower the clock rate of circuits to power on all chip area — **or we have to operate all of them at a slower speed**

  as $P_{dynamic} \sim \alpha \times C \times V^2 \times f \times N$

  A. 0
  B. 1
  C. 2
  D. 3
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Recap — Take-aways: parallel programming

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- Processor behaviors are non-deterministic
  - You cannot predict which processor is going faster
  - You cannot predict when OS is going to schedule your thread
  - You cannot predict when the processor is going to schedule an instruction
- Cache consistency is hard to support
- **Even if we can address all programming challenges, multi-core performance has stopped to scale due to the Dark Silicon problem**

Challenges and state-of-the-art work-arounds in the dark silicon era

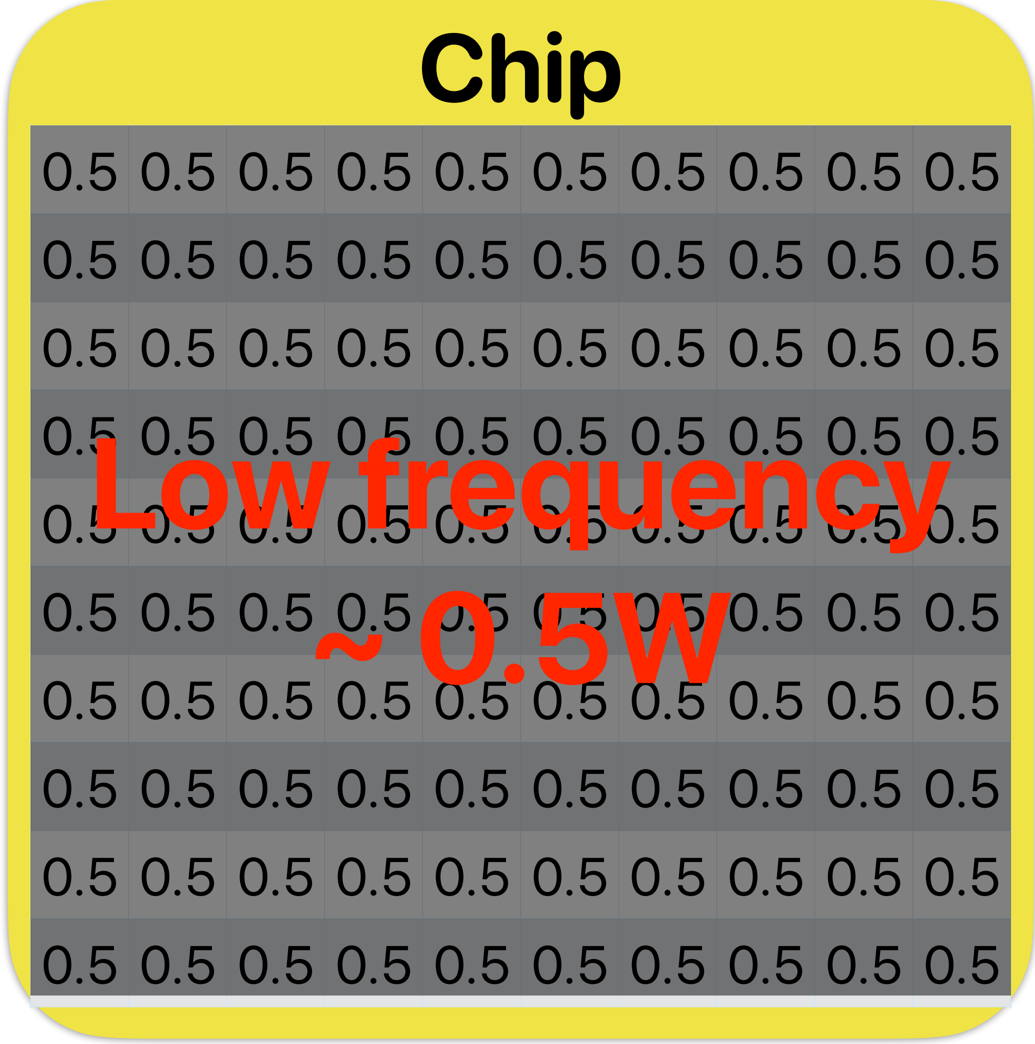
Dark silicon problem

- We are given the same area, twice amount of transistors
- We are given the same power budget
- We are given a certain applications
- How can we maximize the performance for these applications within the given constraints?

How to optimize CMPs under dark silicon problems?

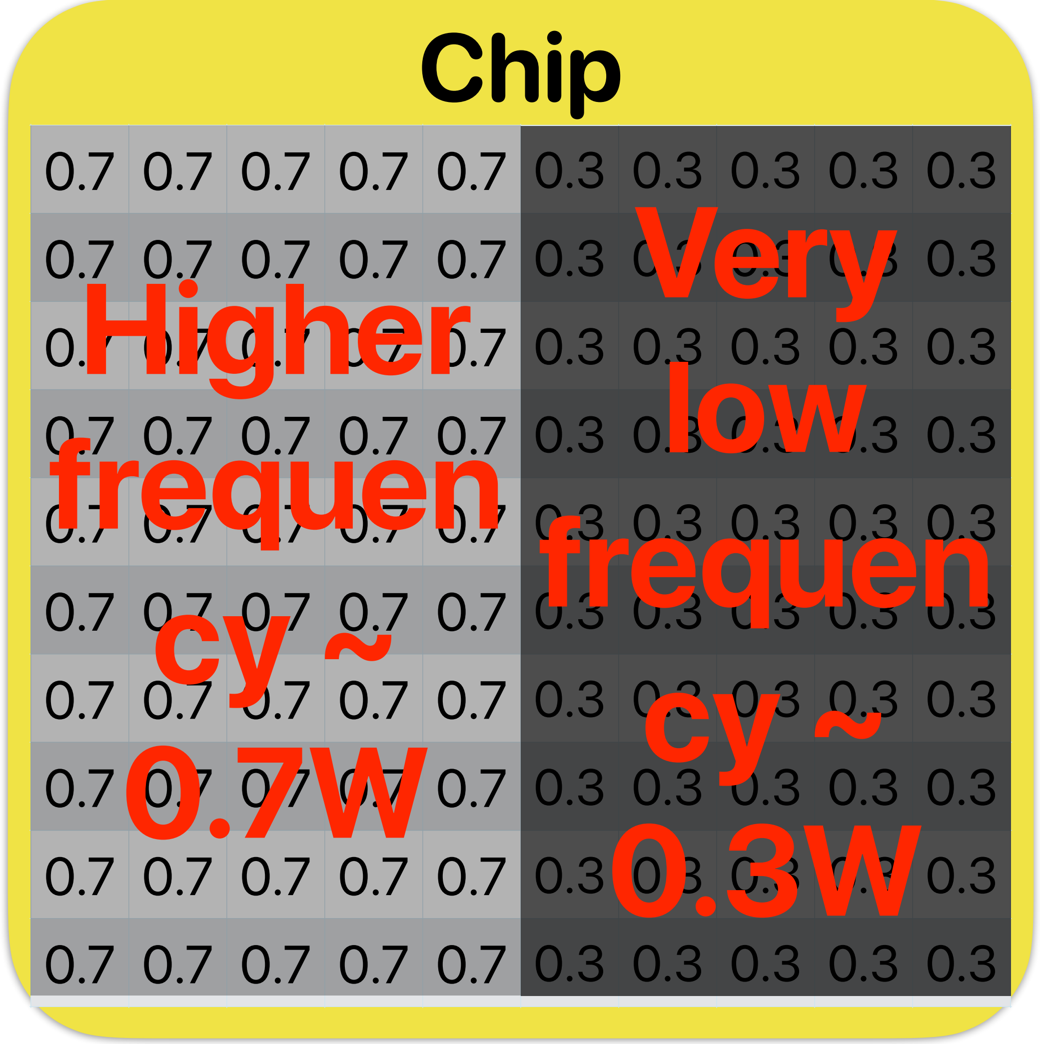
Given the same power budget, maximize the efficiency per chip

Slowdown all of them



=50W!

Some faster, some slower



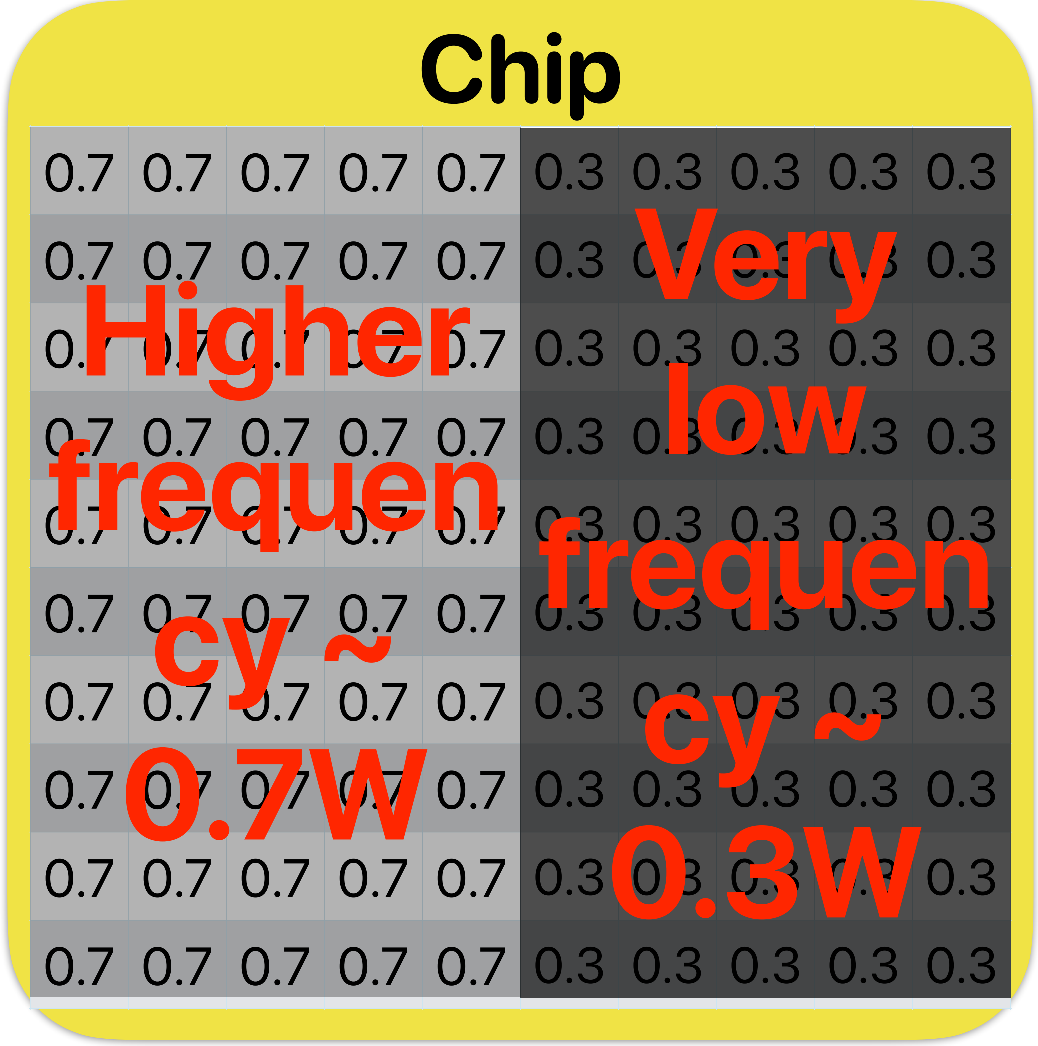
=50W!

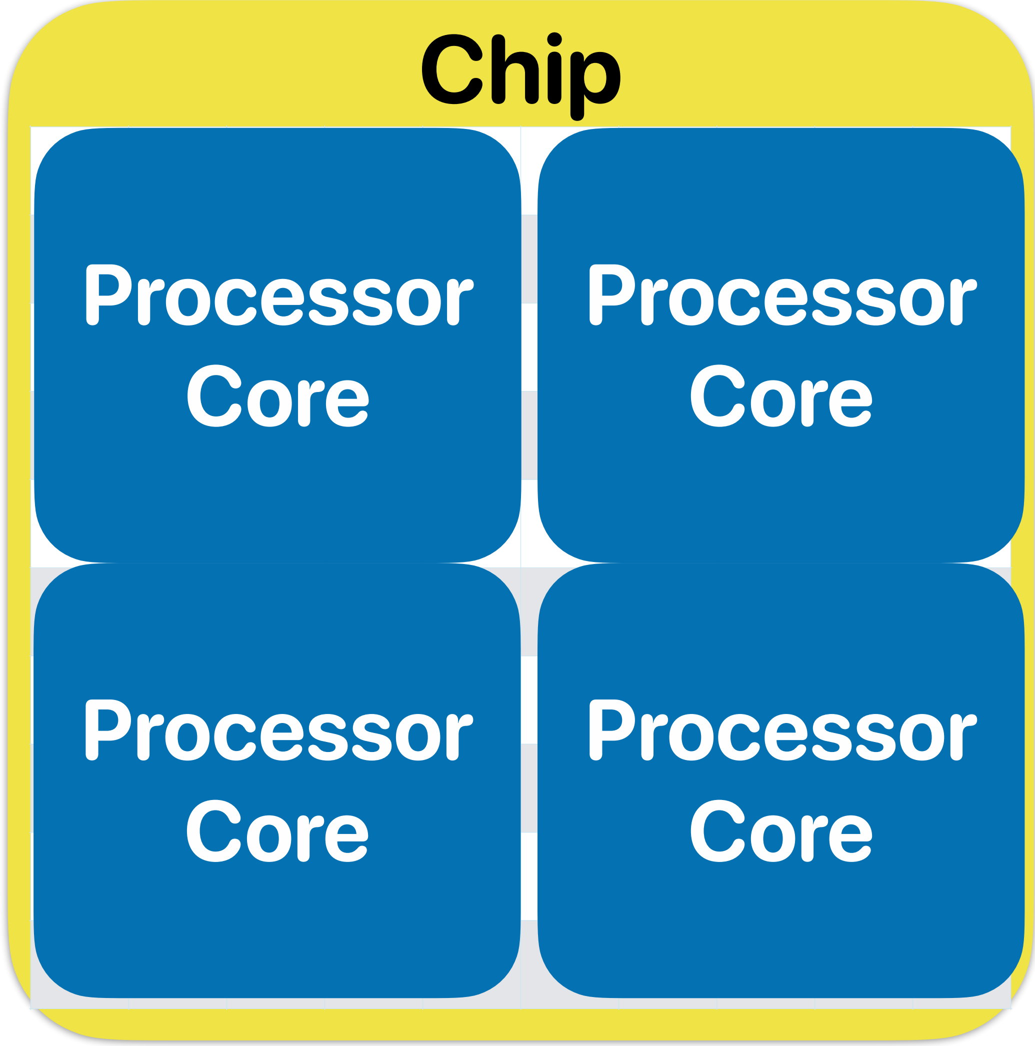
Some at top speed, some are not functioning

=50W!

Given the same power budget, maximize the efficiency per chip

Some faster, some slower,
or even turn off some

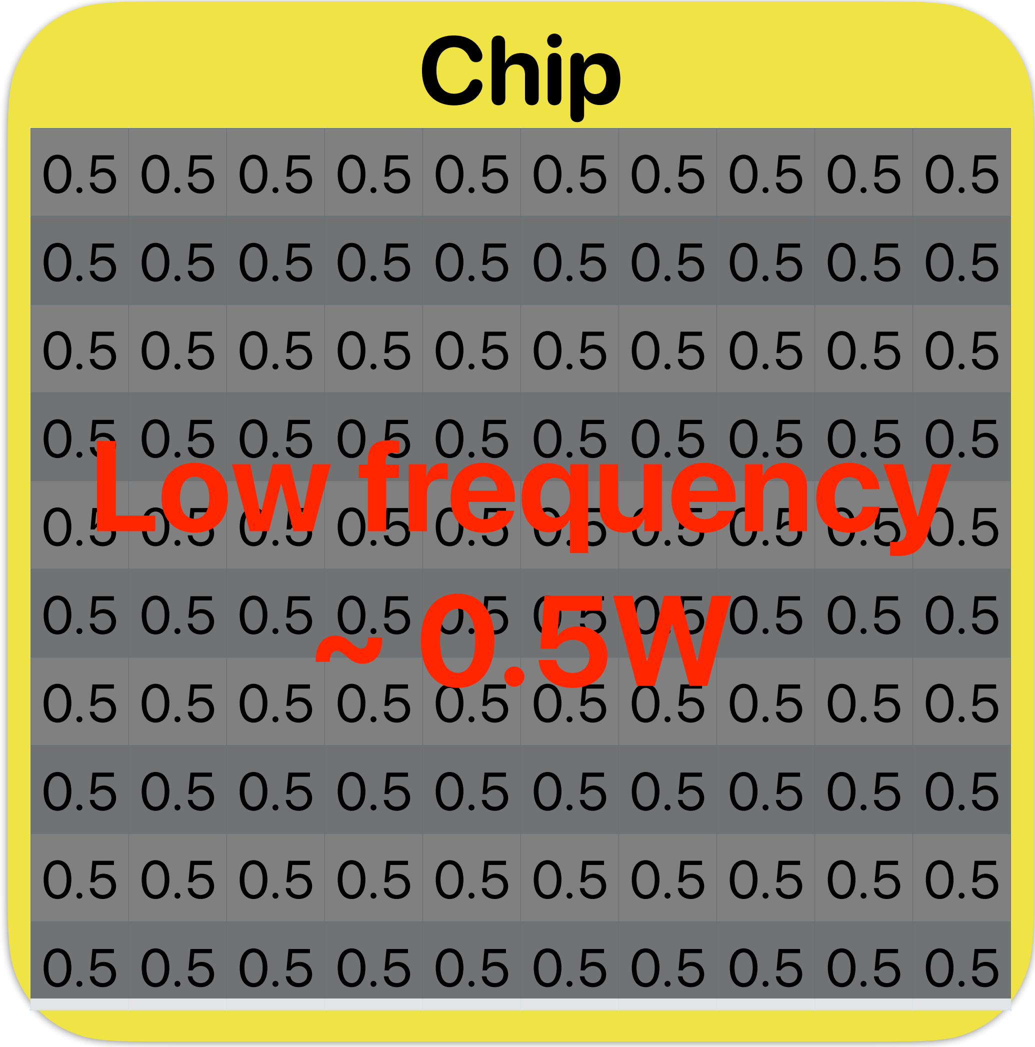


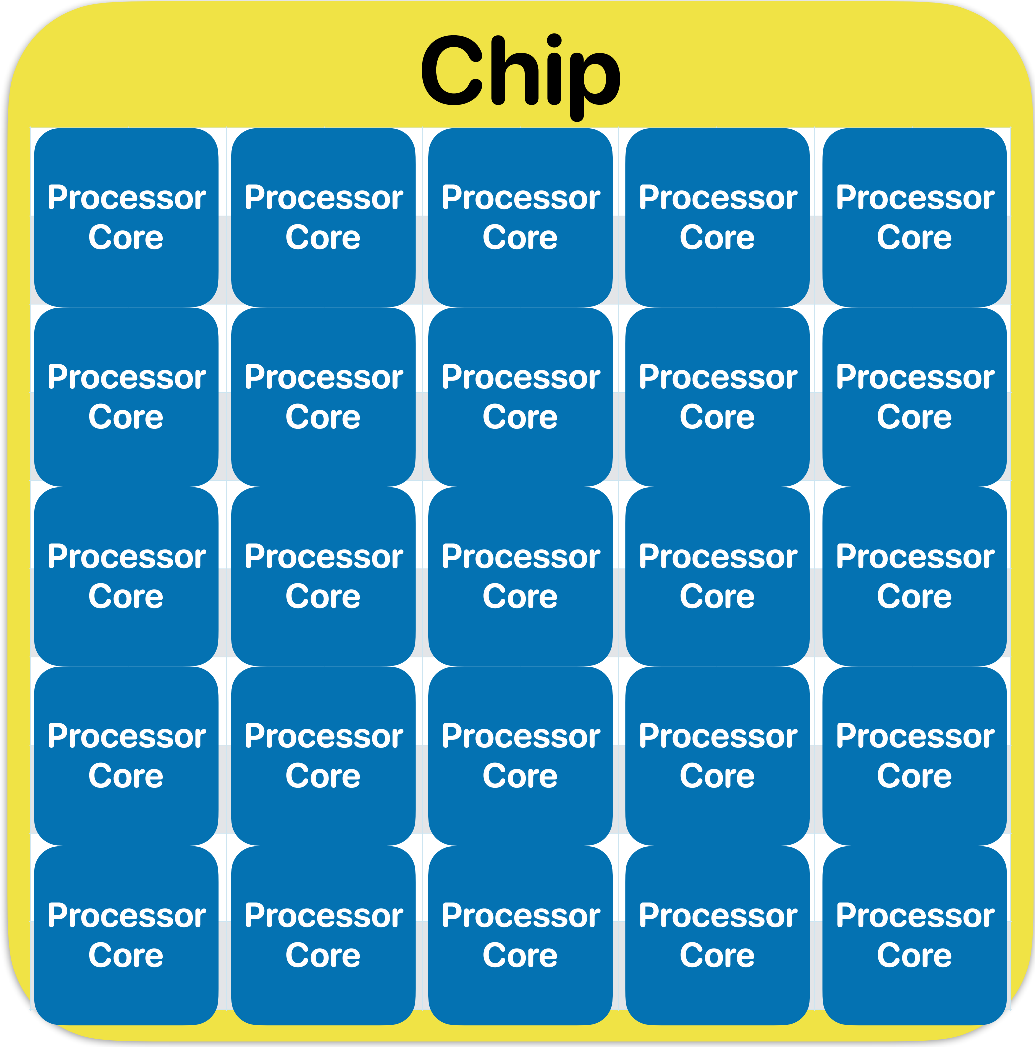


Aggressively adjust the frequency of processor cores — If only one program is compute-intensive, having it running at full speed, others at lower frequency or turning off

Given the same power budget, maximize the efficiency per chip

Slowdown all of them





All of them are smaller, simpler, lower-frequency, lower-power cores

=50W!

Modern processor's frequency

```
Socket(s):                       1
NUMA node(s):                    1
Vendor ID:                       GenuineIntel
CPU family:                      6
Model:                           151
Model name:                      12th Gen Intel(R) Core(TM) i7-12700KF
Stepping:                        2
CPU MHz:                         1226.409
CPU max MHz:                     5000.0000
CPU min MHz:                     800.0000
BogoMIPS:                        7219.20
```

	i7-12700K
	Intel 7
	$450.00 - $460.00
	PC/Client/Tablet, Workstation

CPU Specifications

Total Cores	12
# of Performance-cores	8
# of Efficient-cores	4
Total Threads	20
Max Turbo Frequency	5.00 GHz
Intel® Turbo Boost Max Technology 3.0 Frequency [1]	5.00 GHz
Performance-core Max Turbo Frequency	4.90 GHz
Efficient-core Max Turbo Frequency	3.80 GHz
Performance-core Base Frequency	3.60 GHz
Efficient-core Base Frequency	2.70 GHz
Cache	25 MB Intel® Smart Cache

https://www.intel.com/content/www/us/en/products/sku/134594/intel-core-i712700k-processor-25m-cache-up-to-5-00-ghz/specifications.html

Modern processor's frequency

```
Architecture:              x86_64
CPU op-mode(s):            32-bit, 64-bit
Byte Order:                Little Endian
Address sizes:             48 bits physical, 48 bits virtual
CPU(s):                    12
On-line CPU(s) list:       0-11
Thread(s) per core:        2
Core(s) per socket:        6
Socket(s):                 1
NUMA node(s):              1
Vendor ID:                 AuthenticAMD
CPU family:                25
Model:                     80
Model name:                AMD Ryzen 5 5500
Stepping:                  0
Frequency boost:           enabled
CPU MHz:                   3600.000
CPU max MHz:               3600.0000
CPU min MHz:               1400.0000
```

More cores per chip, slower per core

Export comparison	Intel® Xeon® Platinum 849...	Intel® Xeon® Platinum 846...	Intel® Xeon® Gold 6448H ...	Intel® Xeon® Platinum 844...	Intel® Xeon® Gold 6434H ...
Total Cores	60	48	32	16	8
Total Threads	120	96	64	32	16
Max Turbo Frequency	3.50 GHz	3.80 GHz	4.10 GHz	4.00 GHz	4.10 GHz
Processor Base Frequency	1.90 GHz	2.10 GHz	2.40 GHz	2.90 GHz	3.70 GHz
Cache	112.5 MB	105 MB	60 MB	45 MB	22.5 MB
Intel® UPI Speed	16 GT/s	16 GT/s	16 GT/s	16 GT/s	16 GT/s
Max # of UPI Links	4	4	3	4	3
TDP	350 W	330 W	250 W	270 W	195 W

https://ark.intel.com/content/www/us/en/ark/products/series/228622/4th-generation-intel-xeon-scalable-processors.html

Xeon Phi

Essentials

Product Collection	Intel® Xeon Phi™ 72x5 Processor Family
Code Name	Products formerly Knights Mill
Vertical Segment	Server
Processor Number	7295
Off Roadmap	No
Status	Launched
Launch Date ?	Q4'17
Lithography ?	14 nm

Performance

# of Cores ?	72
# of Threads ?	72
Processor Base Frequency ?	1.50 GHz
Max Turbo Frequency ?	1.60 GHz
Cache ?	36 MB L2 Cache
TDP ?	320 W

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Design options for CMPs

- Consider if we can build a CMP with either (A) 4 powerful cores with aggressive frequency adjustment capabilities or (B) 24 wimpy, slower, narrower-issue cores, please identify the projections that are likely to be true all the time
  ① For single thread applications and if the system has a low number of parallel tasks, A would deliver better performance than B
  ② For single thread applications and if the system has a low number of parallel tasks, B would deliver better energy-efficiency than A
  ③ For multithread applications and if the system has a high number of parallel tasks, B would deliver better performance than A
  ④ For multithread applications and if the system has a high number of parallel tasks, B would deliver better energy-efficiency than A

  A. 0
  B. 1
  C. 2
  D. 3
  E. 4

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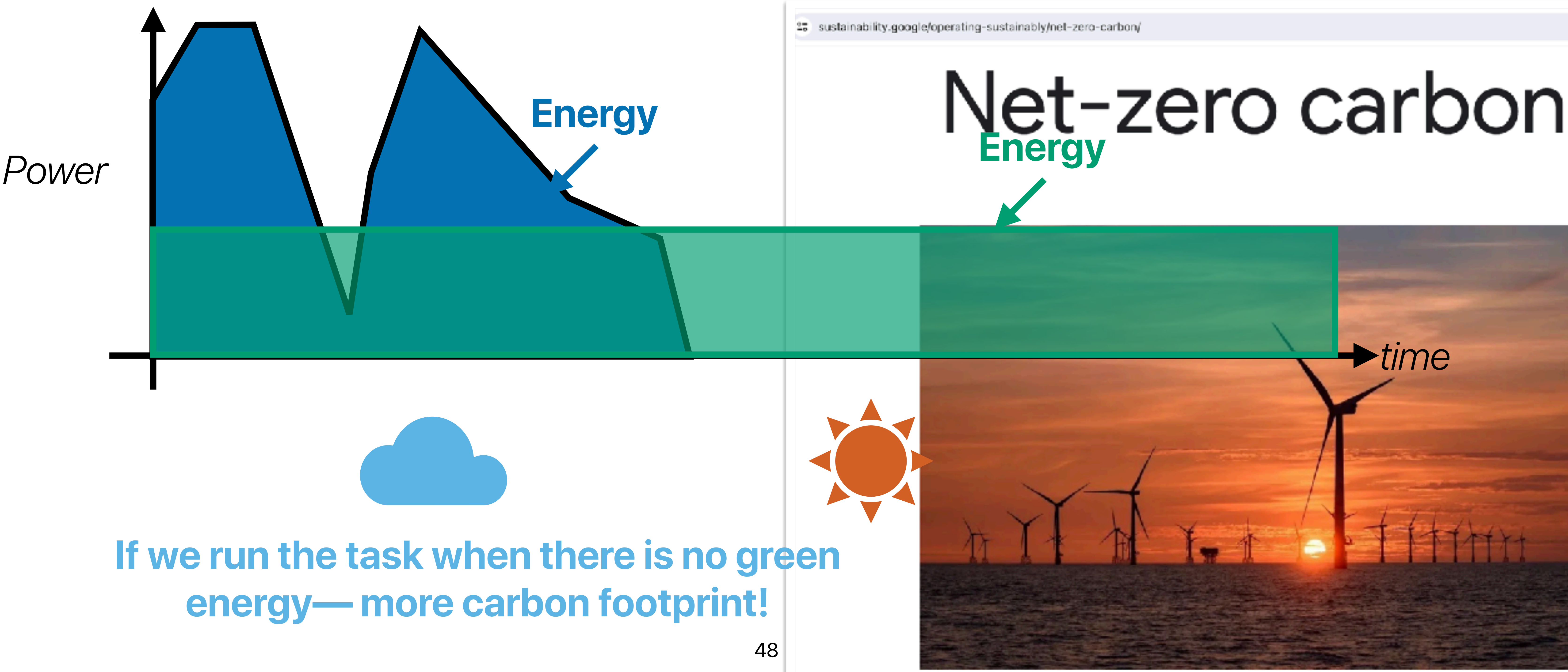
Demo — changing the max frequency and performance

- Change the maximum frequency of the intel processor — you learned how to do this when we discuss programmer's impact on performance
- LIKWID a profiling tool providing power/energy information
  - likwid-perfctr -g ENERGY [command_line]
  - Let's try blockmm and popcount and see what's happening!

Power/Energy/Carbon footprint

The Green can be more if power is not low enough



If we run the task when there is no green energy— more carbon footprint!

Power v.s. Energy

- Power is the direct contributor of “heat”
  - Packaging of the chip
  - Heat dissipation cost
  - Dynamic power
  - Leakage power
- Energy = $Power \times Execution\_Time$
  - The electricity bill and battery life is related to energy!
  - Lower power does not necessary means better battery life if the processor slow down the application too much

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D. 3

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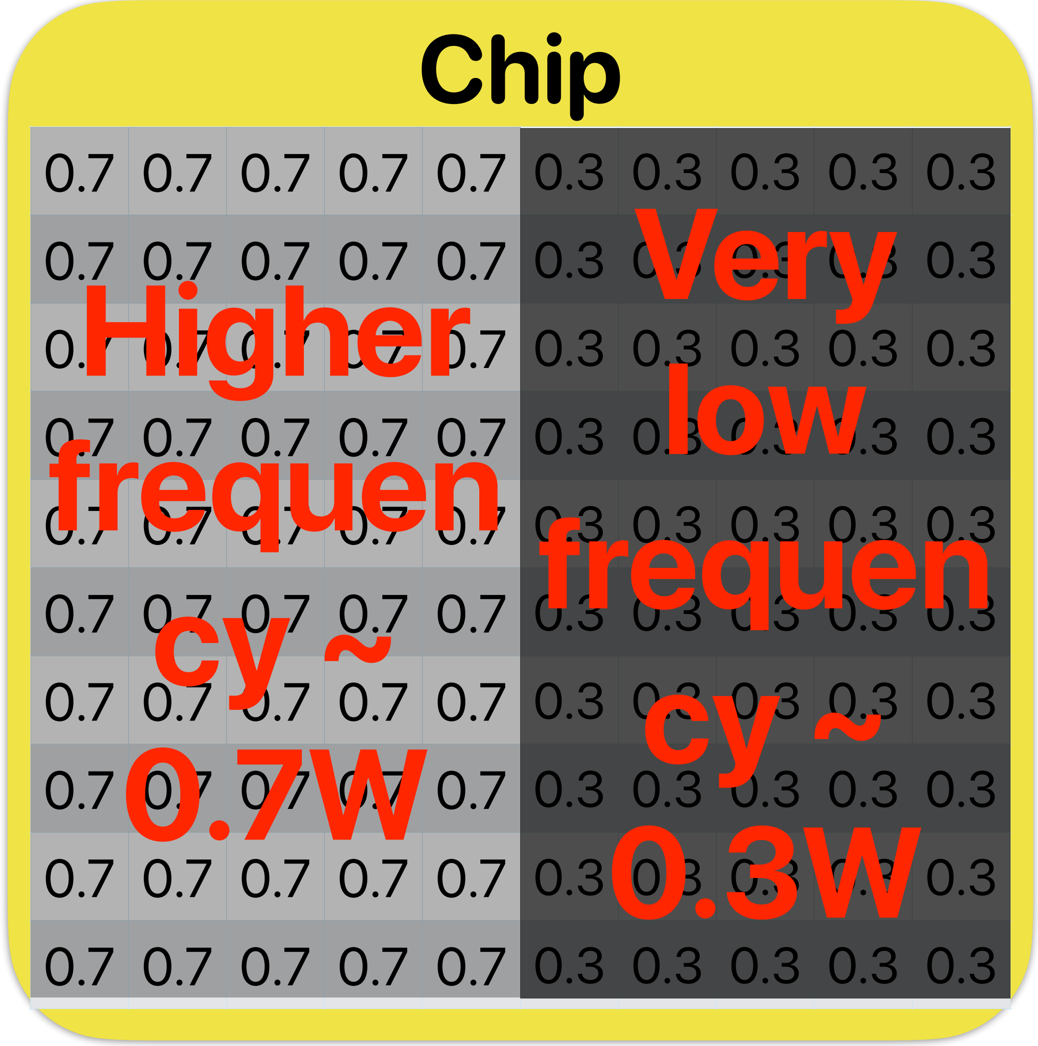
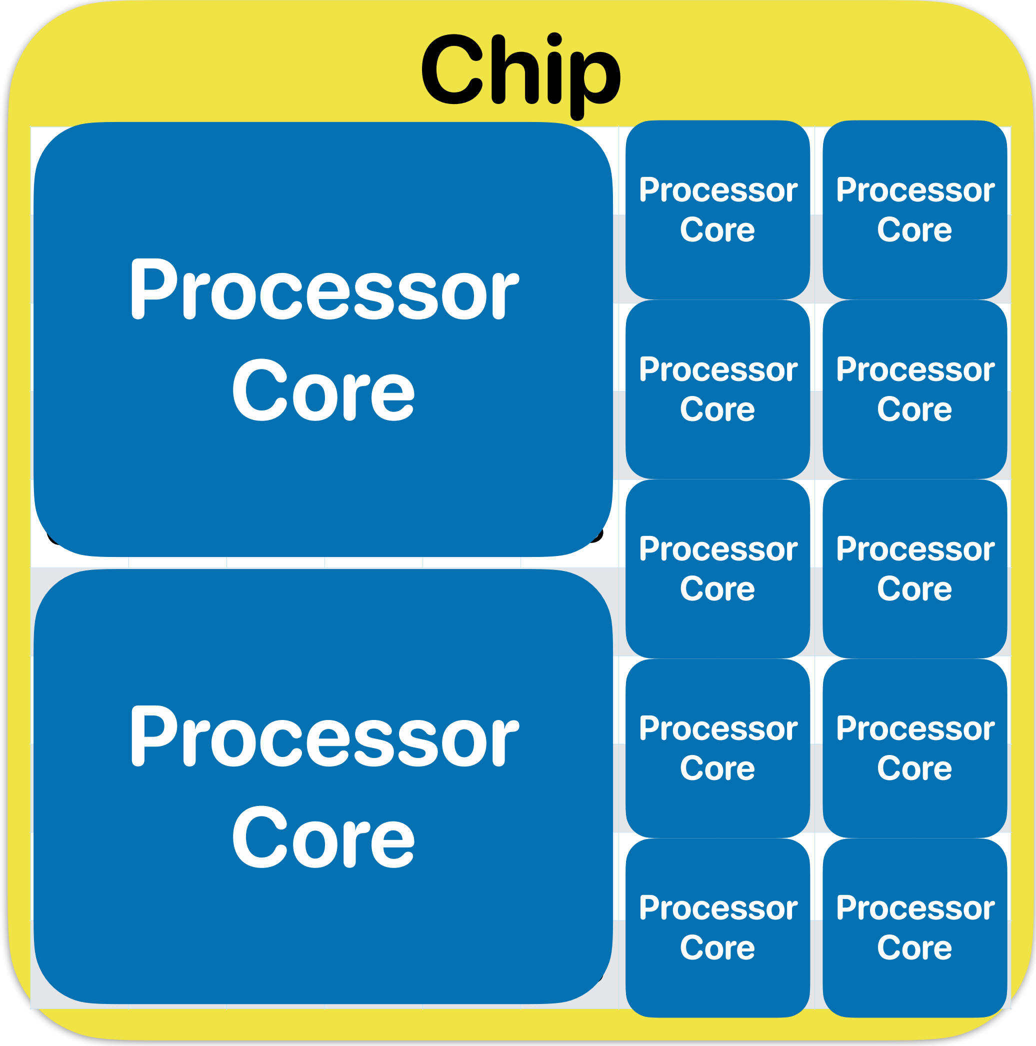
Take-aways: the new golden age of computer architectures

- Challenges and SOTA CMP designs in the dark silicon era
  - Aggressive dynamic frequency/voltage scaling on CMP to accommodate the demand of **latency**-sensitive, parallelism-limited applications, but the area-efficiency of the slower cores is not great
  - Many-core processors improve the **throughput** per-chip through providing massive parallelism where each processing element operates at a lower speed, but not-ideal for latency-sensitive workloads

Given the same power budget, maximize the efficiency per chip

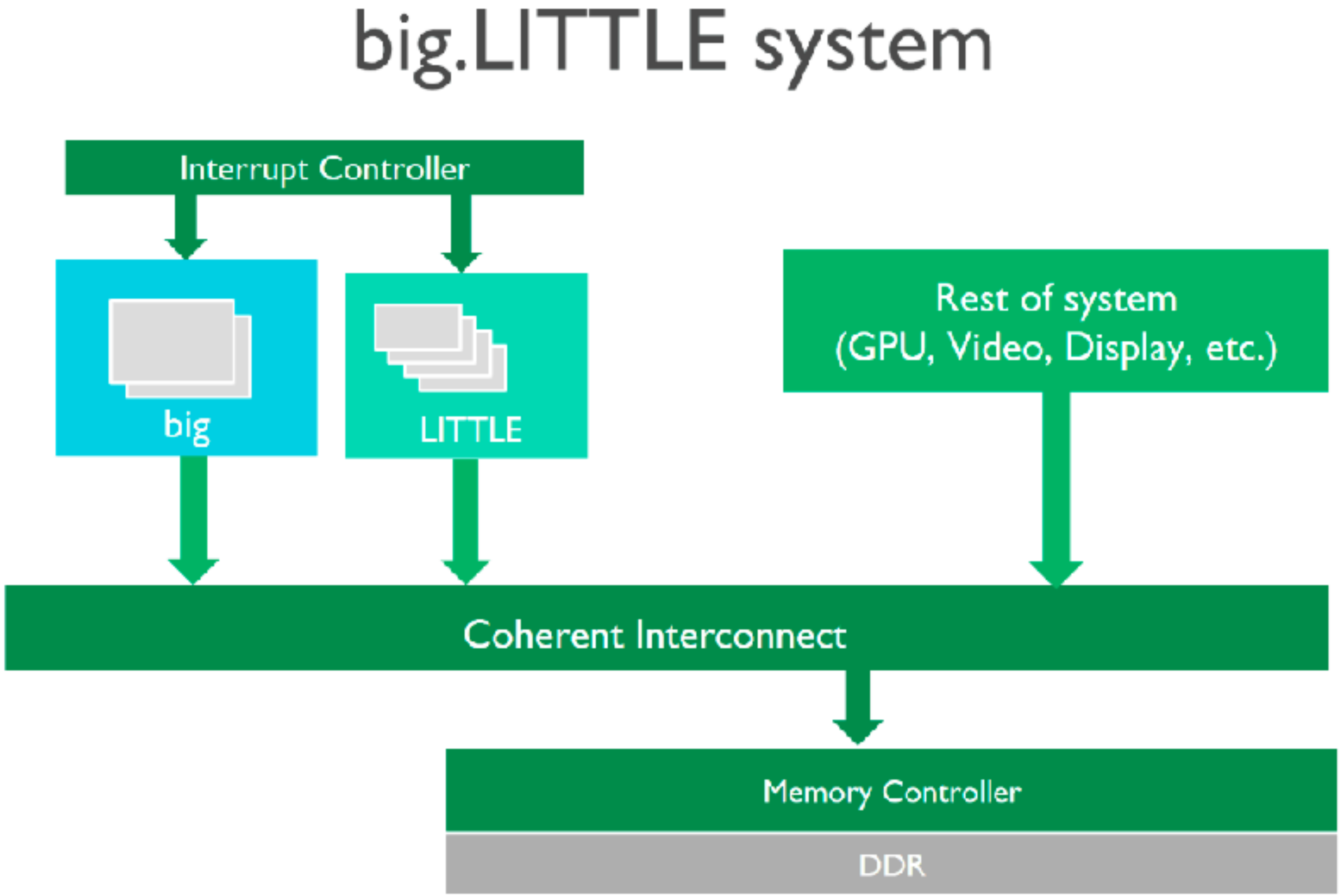
Some faster,
some slower



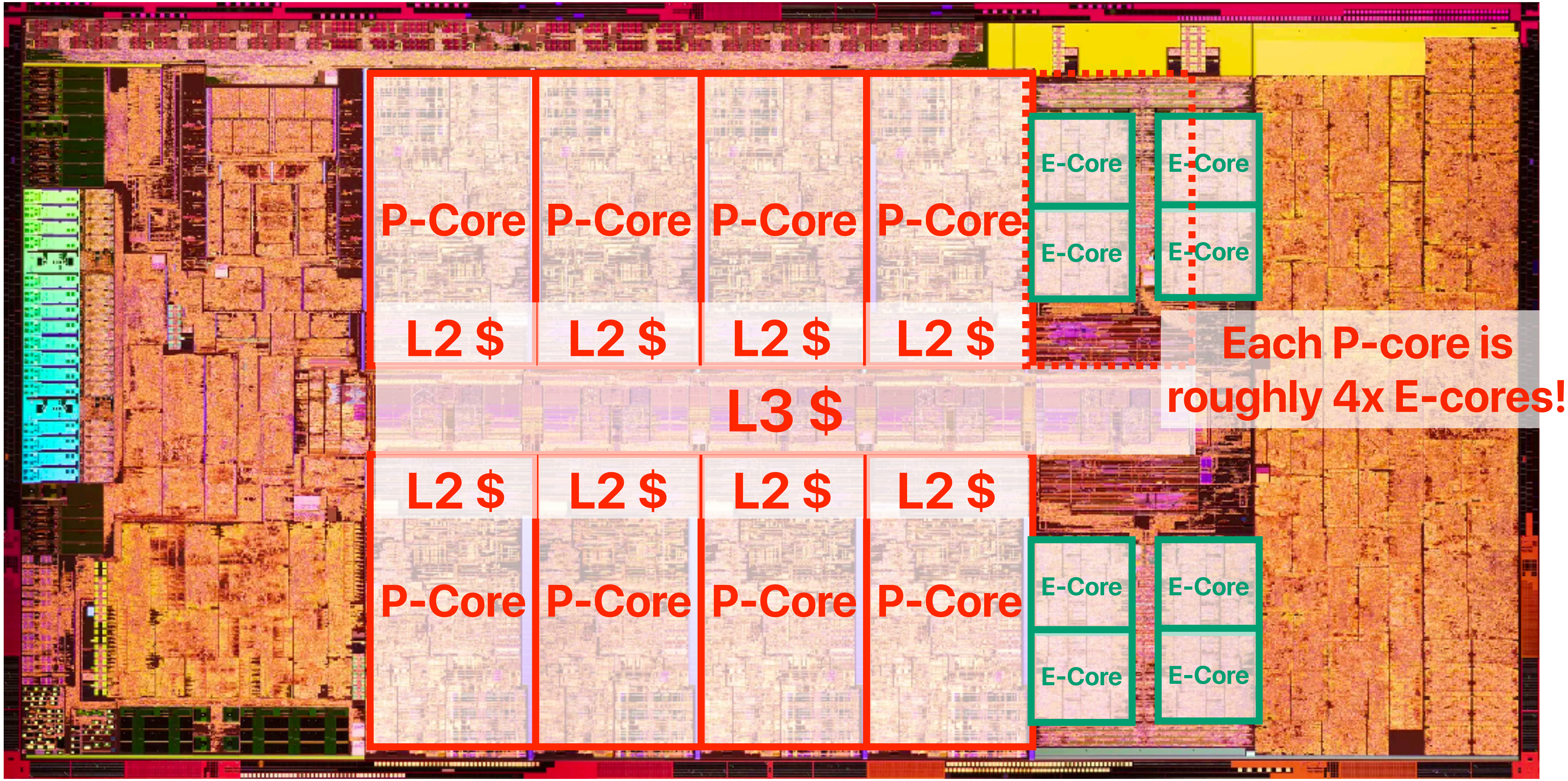


Some powerful cores for latency sensitive workloads, many small cores for highly parallelizable workloads

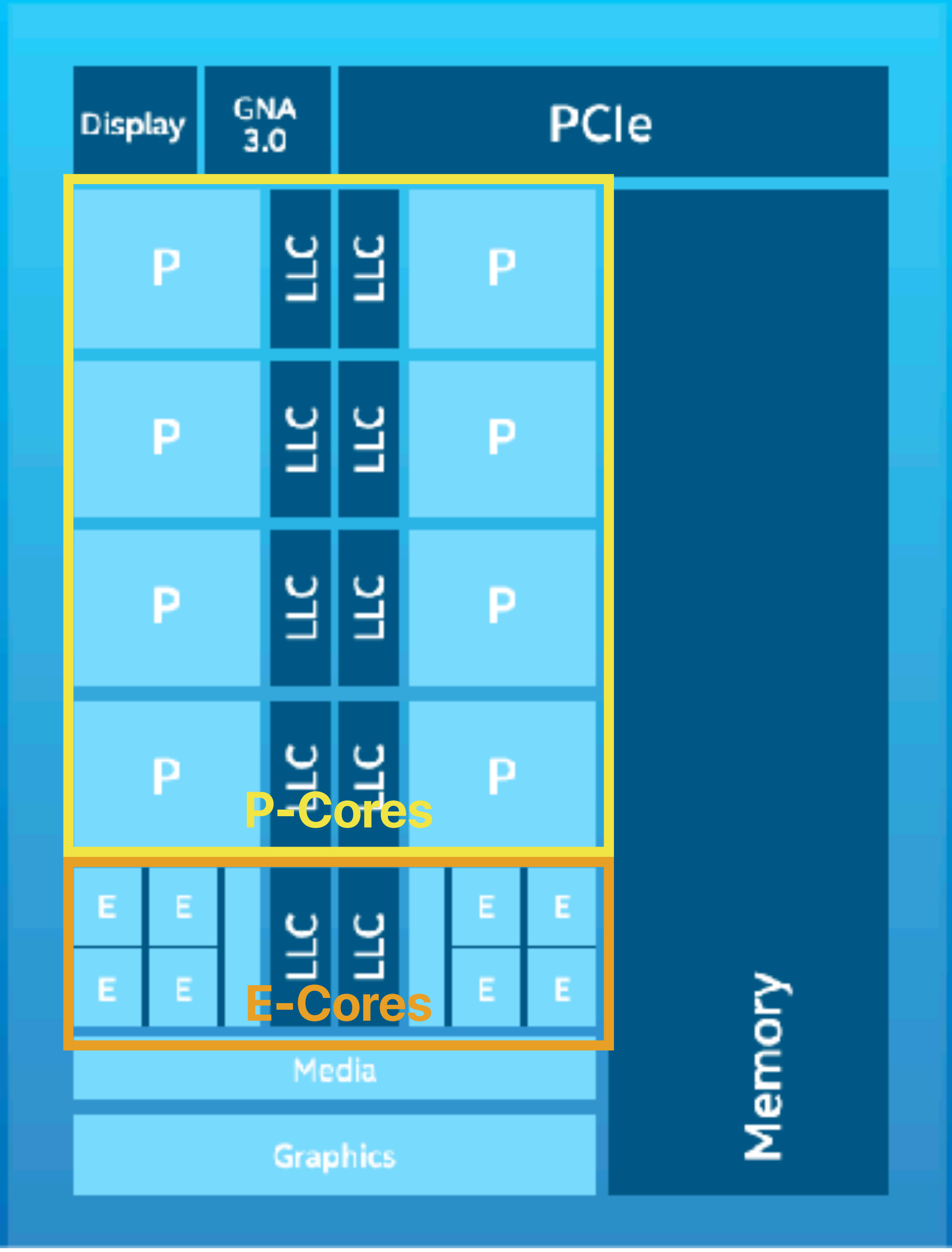
Single ISA heterogeneous CMP in ARM's big.LITTLE architecture



Intel Alder Lake



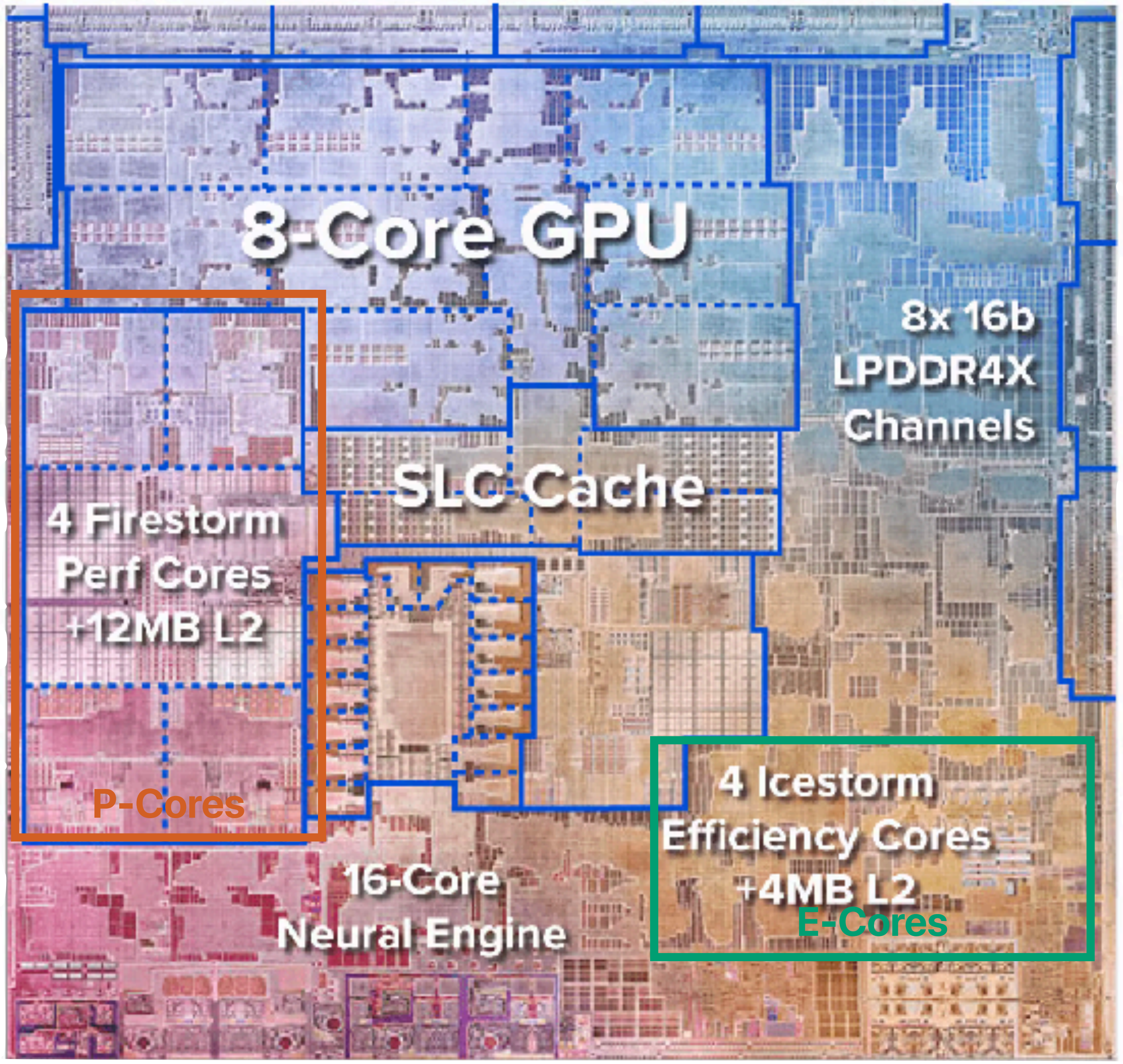
Single ISA heterogeneous CMP in Intel Processors



In a game-changing advance for core performance, 12th Gen Intel® Core™ desktop processors power a revolutionary approach to the x86 architecture. Its Performance-cores—or "P-cores"—are optimized for single & lightly-threaded performance, while its Efficient-cores—or "E-cores"—are optimized for scaling highly-threaded workloads. Intel® Thread Director helps to monitor and analyze performance data in real-time to seamlessly place the right application thread on the right core and optimize performance per watt.[1] That means gamers, creators, and professionals can harness both intelligence and power to enhance the experiences that matter most.

https://download.intel.com/newsroom/2021/client-computing/12th-gen-processor-product-brief.pdf

Single ISA heterogeneous CMP in Apple's M1



https://www.toptal.com/apple/apple-m1-processor-compatibility-overview

https://www.pollev.com/hungweitseng close in 1:30

Single ISA heterogeneous CMP (big.Little)

- Assume we have two processor core architectures and the potential configurations within the same chip area:

	Big Only	Little Only	big.Little
P-Cores: higher performance, but higher power consumption and larger size	4		2
E-Cores: lower performance, but lower power consumption and smaller size		24	12

Please identify the correct expectations of their performance

① For a program with limited thread-level parallelism, **big.Little** would deliver better than or at least the same level of performance as **Little Only**

② For a program with limited thread-level parallelism, **big.Little** would deliver better than or at least the same level of performance as **Big Only**

③ For a program with rich thread-level parallelism, **big.Little** would deliver better or at least the same level of performance than **Little Only**

④ For a program with rich thread-level parallelism, s **big.Little** would deliver better or at least the same level of performance than **big Only**

A. 0
B. 1
C. 2
D. 3
E. 4

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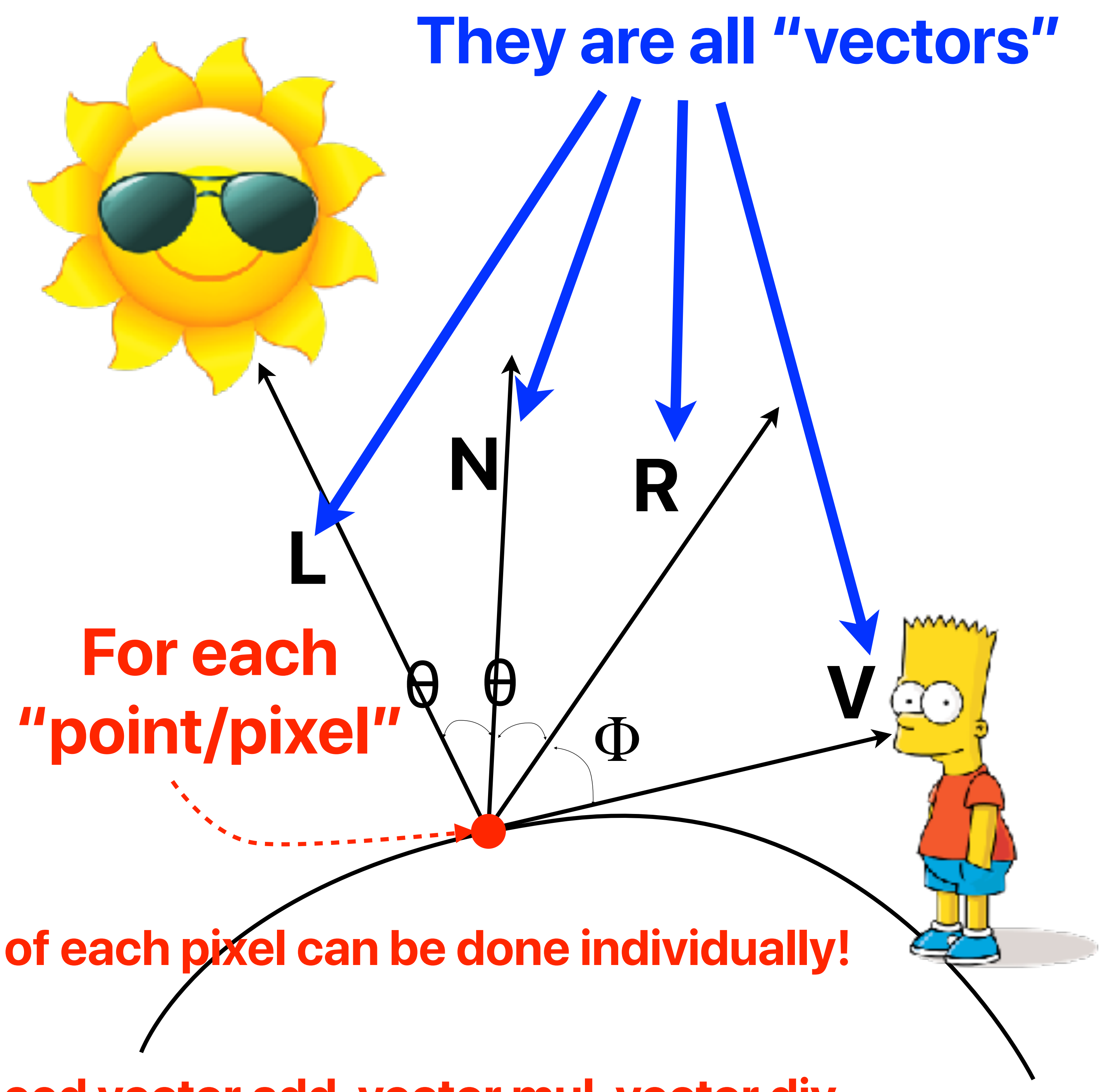
Take-aways: the new golden age of computer architectures

- Challenges and SOTA solutions in the dark silicon era
    - Many-core processors improve the **throughput** per-chip through providing massive parallelism where each processing element operates at a lower speed, but not-ideal for latency-sensitive workloads
    - Aggressive dynamic frequency/voltage scaling on CMP to accommodate the demand of **latency**-sensitive, parallelism-limited applications, but the area-efficiency of the slower cores is not great
    - Single ISA, heterogeneous CMPs (e.g., big.Little cores, Intel/Apple's P-cores/E-cores) find a balance the trade-offs of general-purpose workloads, but won't be ideal if your applications go to either extreme of throughput (massive parallelism) or latency

New type of parallelism

Basic concept of shading



$$I_{amb} = K_{amb} \cdot M_{amb}$$
$$I_{diff} = K_{diff} \cdot M_{diff} \cdot (N \cdot L)$$
$$I_{spec} = K_{spec} \cdot M_{spec} \cdot (R \cdot V)^n$$

$$I_{total} = I_{amb} + I_{diff} + I_{spec}$$

```
void main(void)
{
   // normalize vectors after interpolation
   vec3 L = normalize(o_toLight);
   vec3 V = normalize(o_toCamera);
   vec3 N = normalize(o_normal);

   // get Blinn-Phong reflectance components
   float Iamb = ambientLighting();
   float Idif = diffuseLighting(N, L);
   float Ispe = specularLighting(N, L, V);

   // diffuse color of the object from texture
   vec3 diffuseColor = texture(u_diffuseTexture, o_texcoords)

   // combination of all components and diffuse color of the
   resultingColor.xyz = diffuseColor * (Iamb + Idif + Ispe);
   resultingColor.a = 1;
}
```

The work of each pixel can be done individually!

We only need vector add, vector mul, vector div.

https://www.pollev.com/hungweitseng close in 1:30

What GPU architectures should look like?

- Based on the original target of GPU design, what do you think would be reasonable design decisions that GPU architectures make?
  ① The architecture should render pixels as fast as possible
  ② The architecture does not need a branch unit
  ③ The architecture should contain an array of powerful ALUs and functional units that can perform a rich set of operations
  ④ The architecture should offer very large bandwidth of memory accesses
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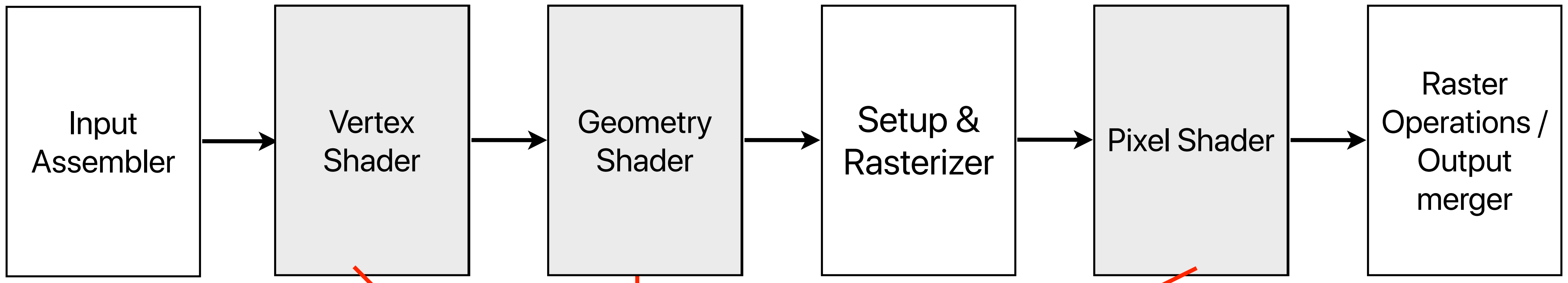
GPU (Graphics Processing Unit)

- Originally for displaying images
- HD video: 1920*1080 pixels * 60 frames per second
  - Therefore, GPU is not latency-oriented by design!
  - Even for 120 frames, you still have 8ms latency to get everything done!
- Graphics processing pipeline

1 GHz can give you 8000000 cycles!!!



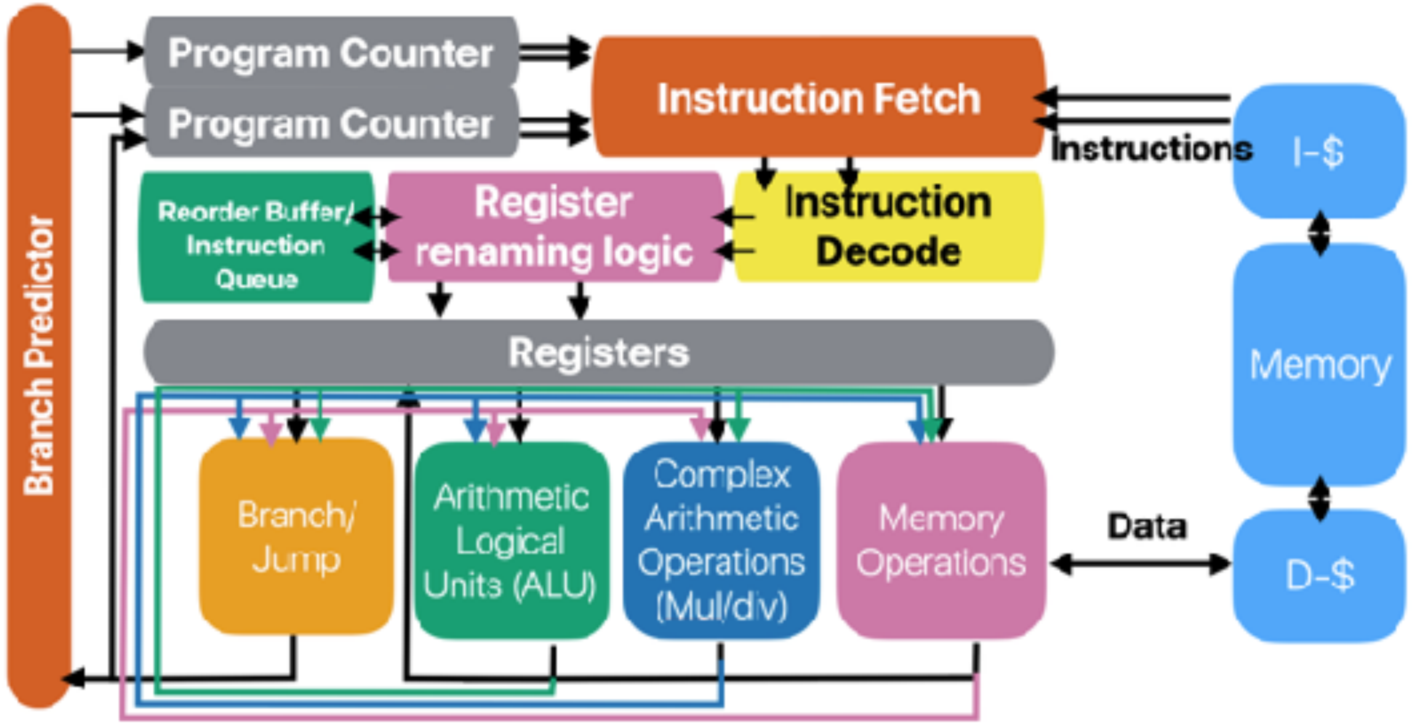
These shaders need to be "programmable" to apply different rendering effects/algorithms (Phong shading, Gouraud shading, and etc...)

Parallelism in modern computers

- Instruction-level parallelism — perform various, independent instructions simultaneously
  - Pipeline
  - OoO/Superscalar
- Thread-level parallelism — perform independent computation streams (composed of many instructions or SIMD instructions)
  - Multicore/SMT processors

How well does vector processing map to super "scalar" processors

```
for(uint64_t i = 0; i < m; i++) {
    result = 0;
    for(uint64_t j = 0; j < n; j++) {
        result += matrix[i][j]*vector[j];
    }
    output[i] = result;
}
```



Assume we have a 5-issue INT, 3-load, 4-store pipeline like Alder Lake

Performance is very limited by both the memory bandwidth and available functional units

```
load vector[0]          load matrix[0][1]          load vector[3]          load matrix[0][4]
load matrix[0][0]       load vector[2]             load matrix[0][3]       load vector[5]
load vector[1]          load matrix[0][2]          load vector[4]          load matrix[0][5]

                        mul matrix[0][0], vector[0] mul matrix[0][1], vector[1] mul matrix[0][3], vector[3]
                                                    mul matrix[0][2], vector[2]
```

Characteristics of vector processing

- Their operations are uniform across all pairs of elements

— Can we have just one instruction to control all pair-wise operations?

- There are very limited vector operations with mathematical meanings

— Can we simplify the processing elements design and make space for more PEs?

- They have very good spatial locality

— Can we have fetch them once & put in wide registers?

```
for(uint64_t i = 0; i < m; i++) {
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    for(uint64_t j = 0; j < n; j++) {
          result += matrix[i][j]*vector[j];
    }
     output[i] = result;
}
```

```
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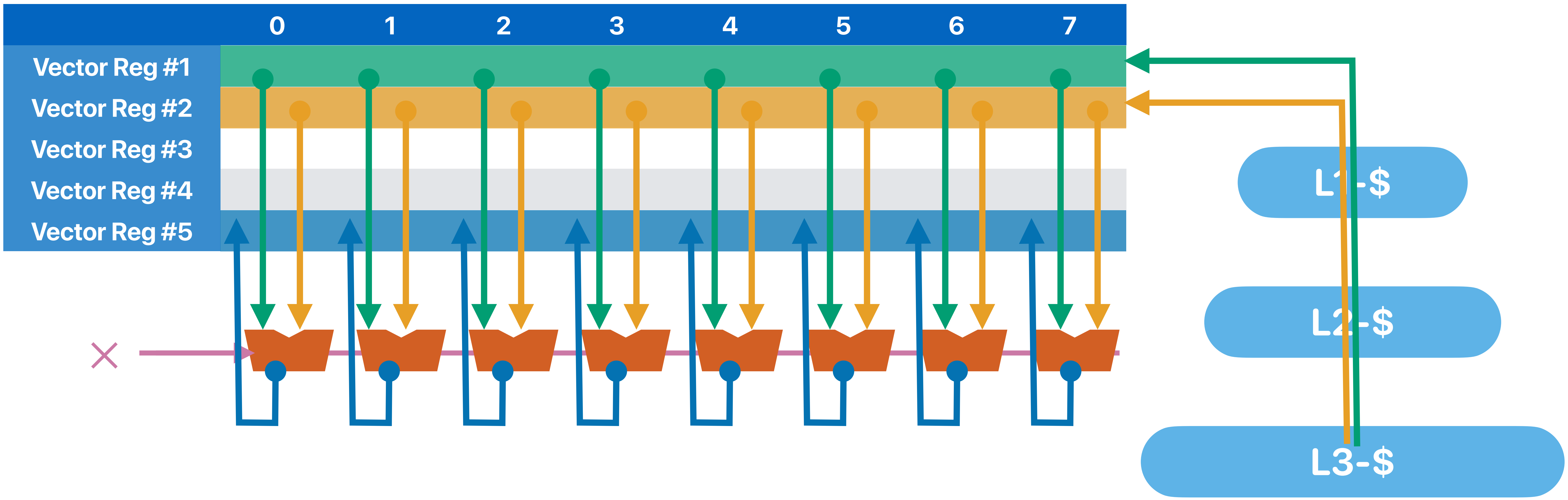
mul matrix[0][0], vector[0]   mul matrix[0][1], vector[1]   mul matrix[0][3], vector[3]
                              mul matrix[0][2], vector[2]
```

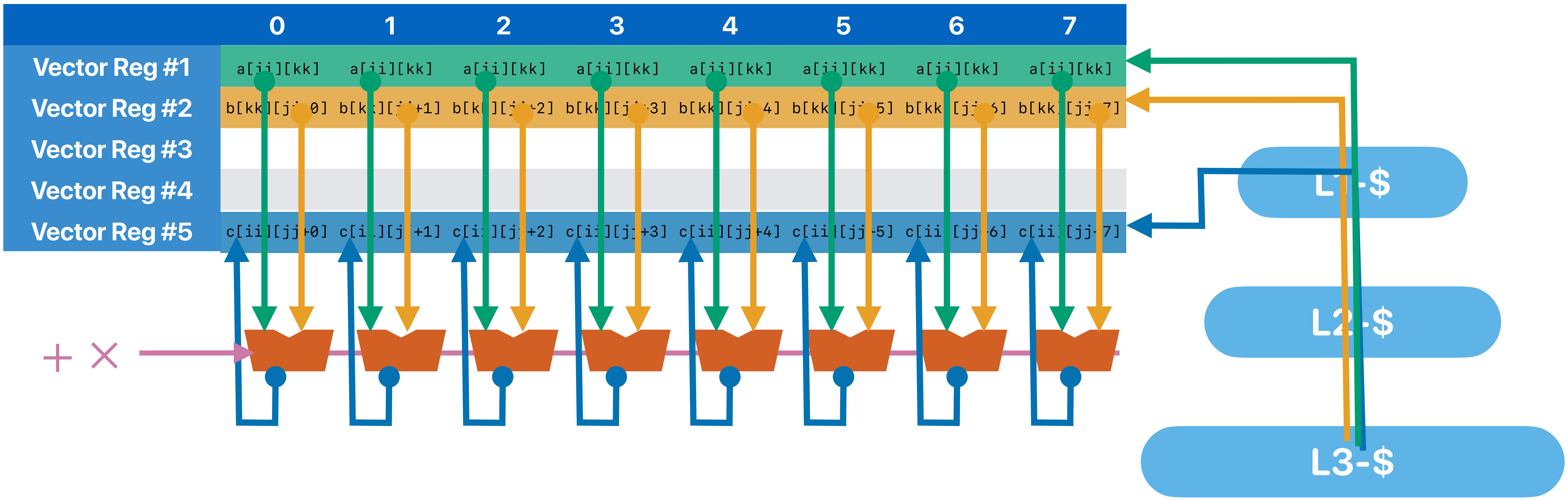
Vector processing architecture



Vectorized matrix multiplications

```
#define VECTOR_WIDTH 8 // VECTOR_WIDTH is 8 as we're using AVX-512
void vector_blockmm(double **a, double **b, double **c, \
uint64_t tile_size) {
  int i,j,k, ii, jj, kk, x;
  __m512d va, vb, vc;
  for(i = 0; i < ARRAY_SIZE; i+=tile_size) {
    for(j = 0; j < ARRAY_SIZE; j+=tile_size) {
      for(k = 0; k < ARRAY_SIZE; k+=tile_size) {
        for(ii = i; ii < i+tile_size; ii++) {
          for(jj = j; jj < j+tile_size; jj+=VECTOR_WIDTH) {
             vc = _mm512_load_pd(&c[ii][jj]); // load c[ii][jj] to c[ii][jj+7] to vc[0-7]
             for(kk = k; kk < k+tile_size; kk++) {
                va = _mm512_broadcastsd_pd(&a[ii][kk]);// load a[ii][kk] to va[0-7]
                vb = _mm512_load_pd(&b[kk][jj]);    // load b[kk][jj] to b[kk][jj+7] to vb[0-7]
                vc = _mm512_add_pd(vc,_mm512_mul_pd(va,vb));    // vc += va * vb
             }
             _mm512_store_pd(&c[ii][jj],vc);       // store vc to c[ii][jj] to c[ii][jj+4]
          }
        }
      }
    }
  }
}
```

Vector processing architecture



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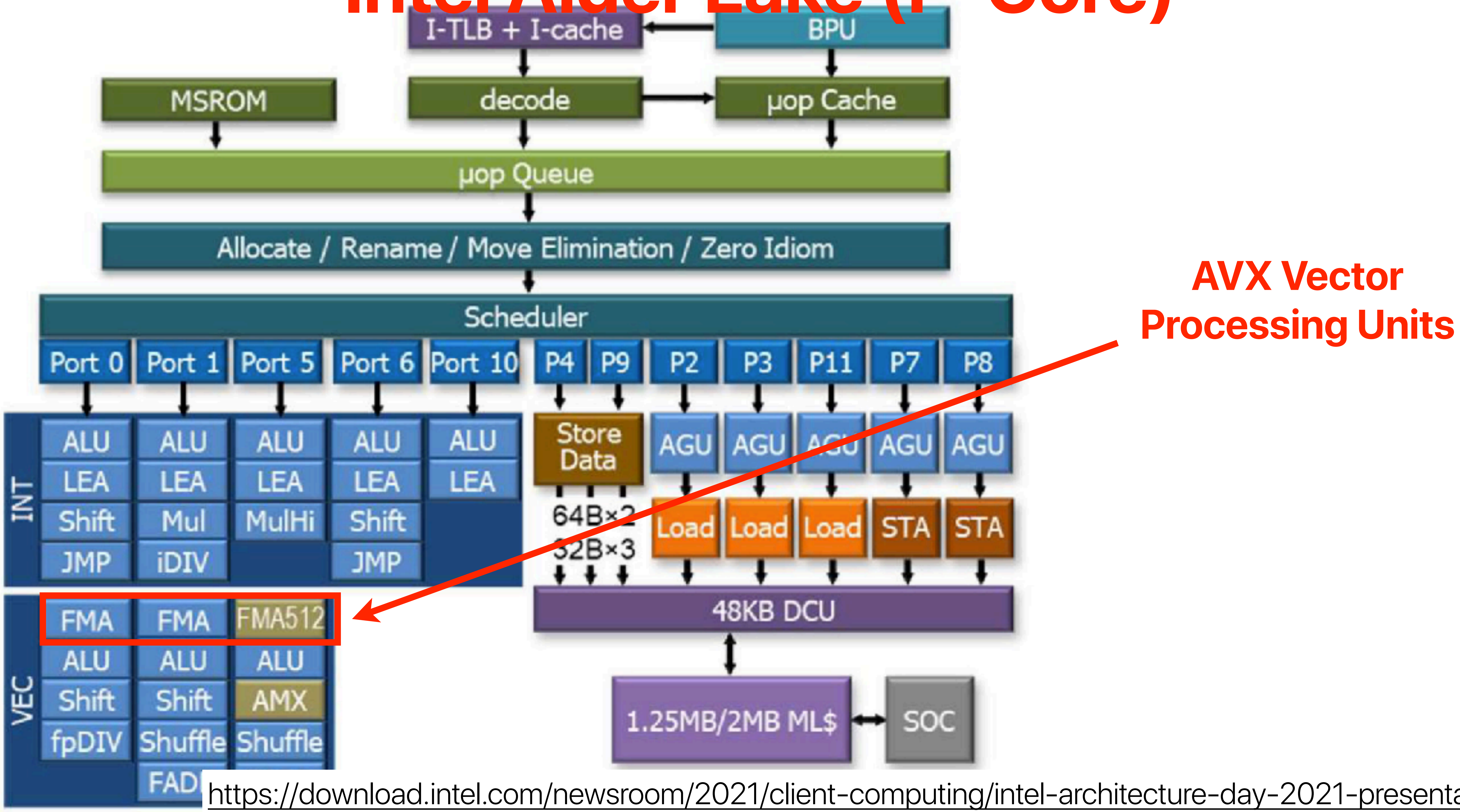
Intel Alder Lake (P-Core)



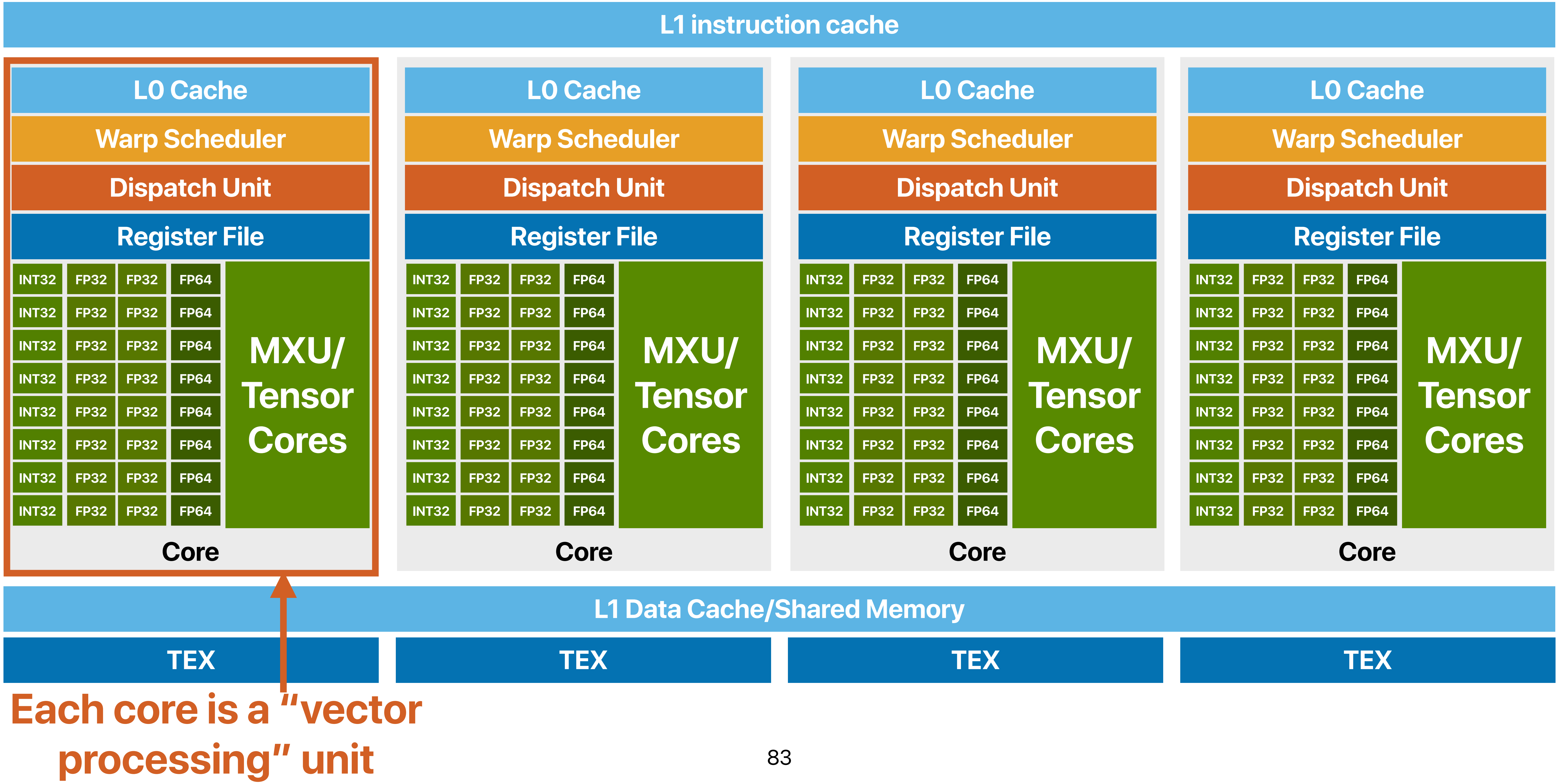
https://download.intel.com/newsroom/2021/client-computing/intel-architecture-day-2021-presentation.pdf

What's the "appropriate" GPU architecture

- Lots of ALUs to process pixels in parallel — 2M pixels in HD resolution, very regular workloads
  - Vector processing model
- Simple operations
  - The ALUs only supports very few instructions
  - Almost no branches
- Deadline driven and throughput-oriented rather than latency oriented
  - High-bandwidth but also "higher-latency" memory
  - ALUs can be slower

GPU also follows the idea of slower, but more!

GPU Architecture



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- Data-level parallelism — perform the same operation on multiple data elements in parallel
  - SIMD instructions
  - Compute within an SM in GPUs
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  - Multicore/SMT processors
  - Compute using multiple SMs in GPUs

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  - Single ISA, heterogeneous CMPs (e.g., big.Little cores, Intel/Apple's P-cores/E-cores) find a balance the trade-offs of general-purpose workloads, but won't be ideal if your applications go to either extreme of throughput (massive parallelism) or latency
- GPU architectures help improve the area-efficiency of many-core designs by trading the processor functionality with more parallelism

Announcements

- Final Review this Thursday during the lecture
- **Course evaluation** started and ends on 9/05/2025 @ 8AM
  - Submit the prove of your participation in course evaluation through Gradescope
  - It can become a full credit reading quiz (it helps to amortize the penalty of another least performing one) — we will drop a total of three now
- **Assignment 5 due on 9/05/2025**
  - The due date is earlier to allow publication of solutions before the deadline
- **Final exam**
  - Closed book, no cheatsheet — the same rules as the midterm


Computer
Science &
Engineering
142
ありがとう
Prof. Usagi